

Physical Layer of Communications in IoT

Internet of Things



CS5055 - 2025I
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SRCwww.itu.int

IEEE sources: IEEE 802.15.7 - Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
Jim Kurose and Keith Ross. 2012. Computer Networking a Top Down Approach. Addison-Wesley, 6th edition.
I. F. Akyildiz et al., Wireless sensor networks: a survey. Computer Networks, 38:4, pp. 393-422, 2002.

Executive Summary

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- **Motivation:** Network implementation is half the work of an IoT implementation.
- **Problem:** Knowledge of Physical Layer Communication protocols and technologies applied to IoT
- **Topics:**
 - Basic concepts review
 - Physical Layer
 - 5G
 - LiFi

Outline

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Introduction

5G and beyond

LiFi

Conclusion

Ejemplo 1: Pulsera Fitness simple (no conectable). La pulsera solo quiere transmitir su ID y nivel de batería a cualquier app que pase cerca, para ser identificada en una tienda o gimnasio. No quiere establecer conexión, solo transmitir un anuncio. Tipo de paquete: ADV_NONCONN_IND.

Scanneable, no conectable.

Ejemplo 2: Termómetro BLE que puede ser leído por una app. El termómetro BLE transmite que está disponible. La app del celular escanea, descubre el termómetro y pide más información (marca, última temperatura), pero no establece conexión permanente. Tipo de paquete: ADV_SCAN_IND.

No conectable, no escaneable.

Ejemplo 3: Reloj inteligente esperando ser conectado. El smartwatch está esperando que el teléfono se conecte para sincronizar pasos y notificaciones. Anuncia que está disponible y acepta conexiones. Tipo de paquete: ADV_IND.

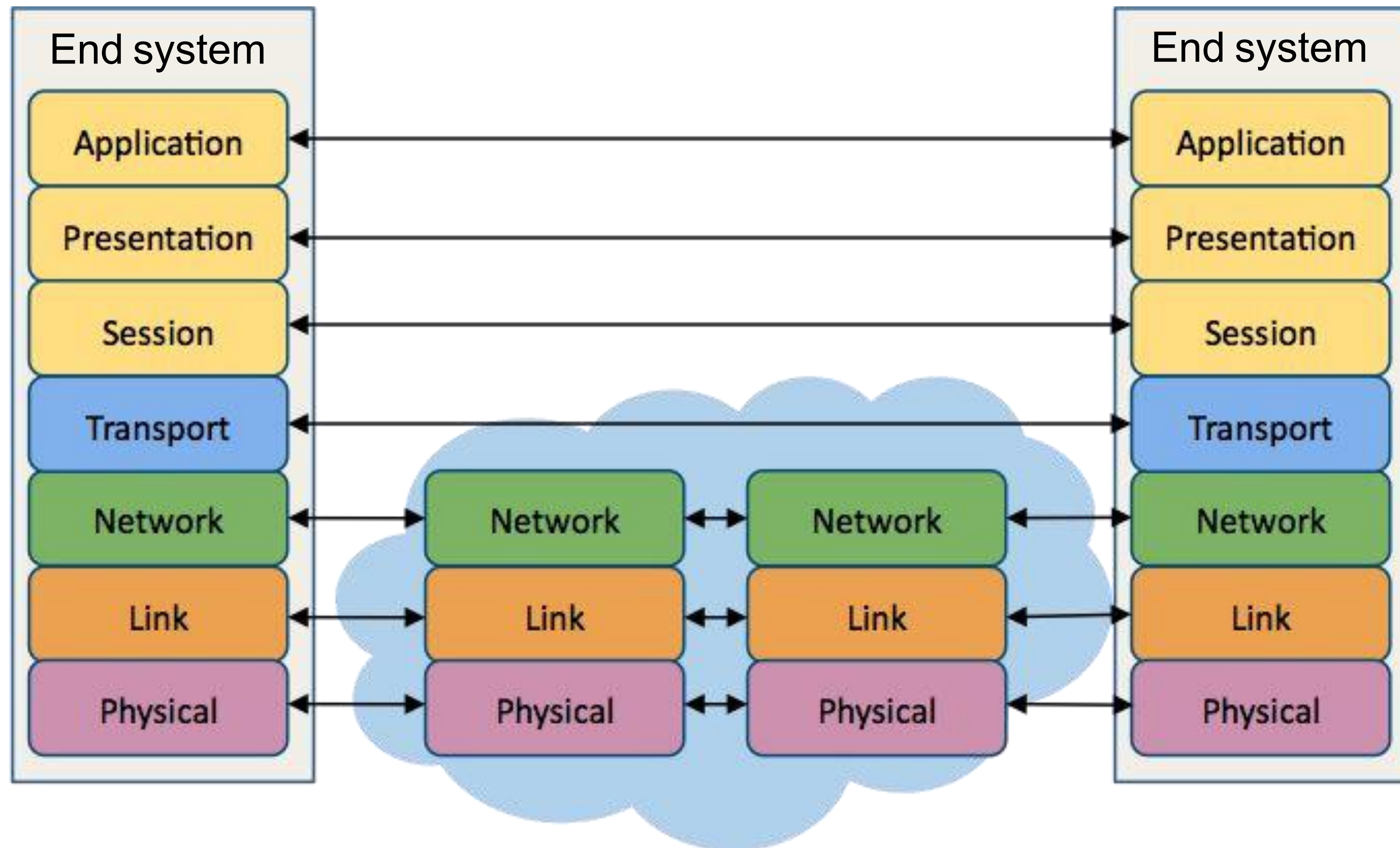
Conectable, dirigido.

Ejemplo 4: Dispositivo emparejado previamente. Los auriculares BLE ya están emparejados con el celular. Ahora el celular los busca directamente y los auriculares hacen directed advertising solo hacia ese celular, para agilizar la reconexión. Tipo de paquete: ADV_DIRECT_IND.

Conectable, scanneable.

Open Systems Interconnection (OSI) Model

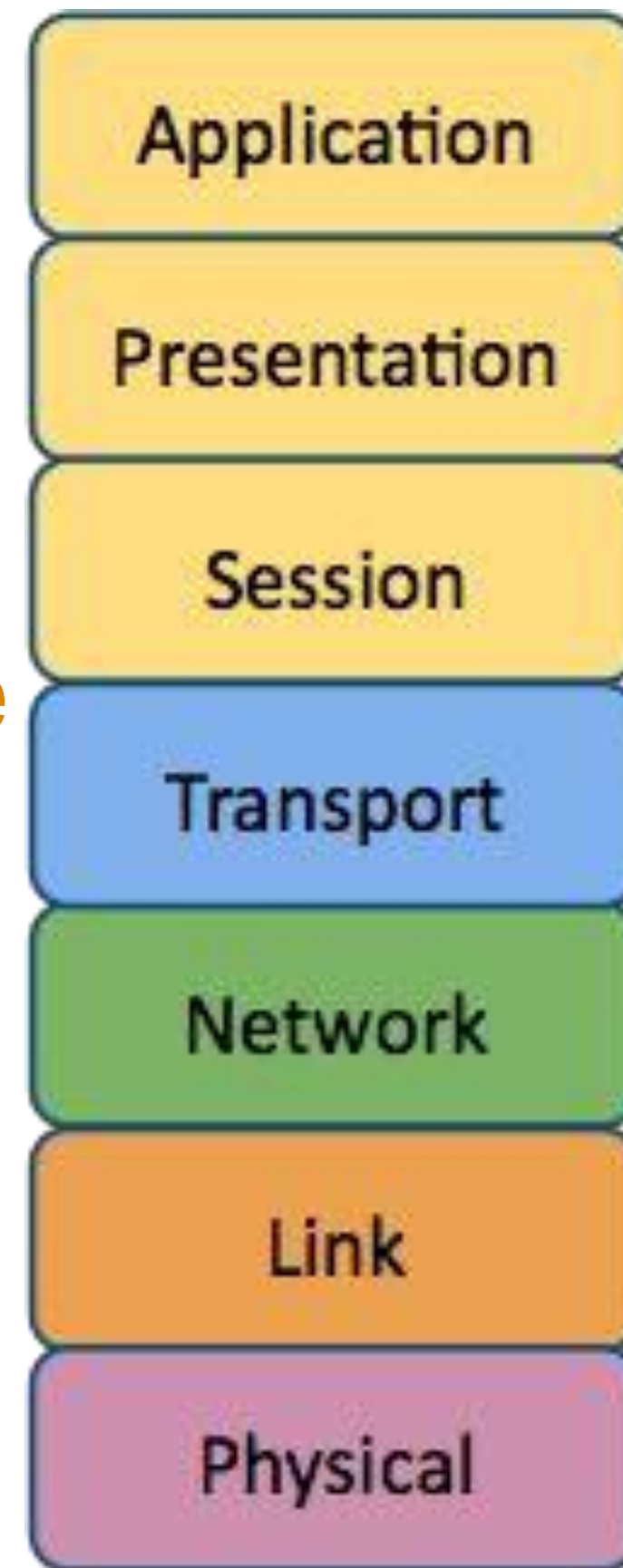
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OSI Model

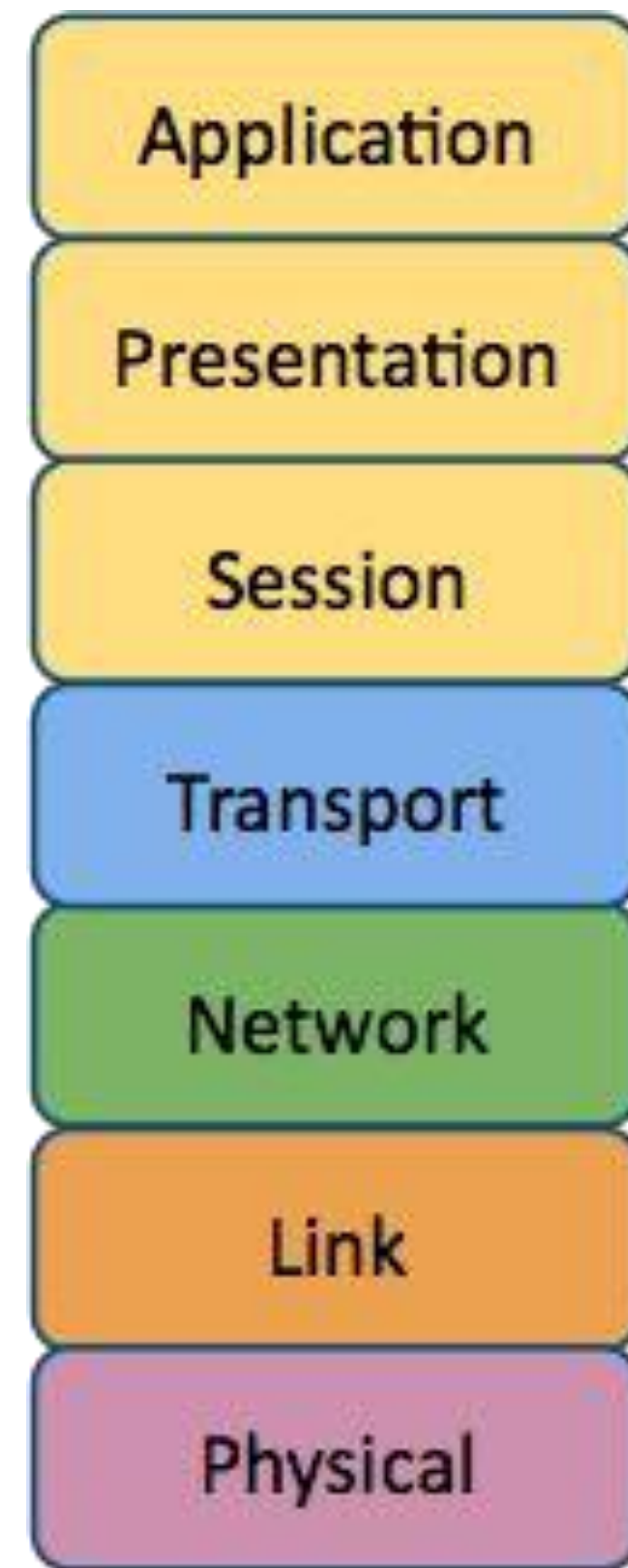
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- **Application:** supporting network applications
 - FTP, SMTP, HTTP
- **Presentation:** allow applications to interpret meaning of data
 - encryption, compression, machine-specific conventions
- **Session:** synchronization, checkpointing, recovery of data exchange
- **Transport:** process-process data transfer
 - TCP, UDP
- **Network:** routing of datagrams from source to destination
 - IP, routing protocols
- **Link:** data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi)
- **Physical:** bits “on the wire”

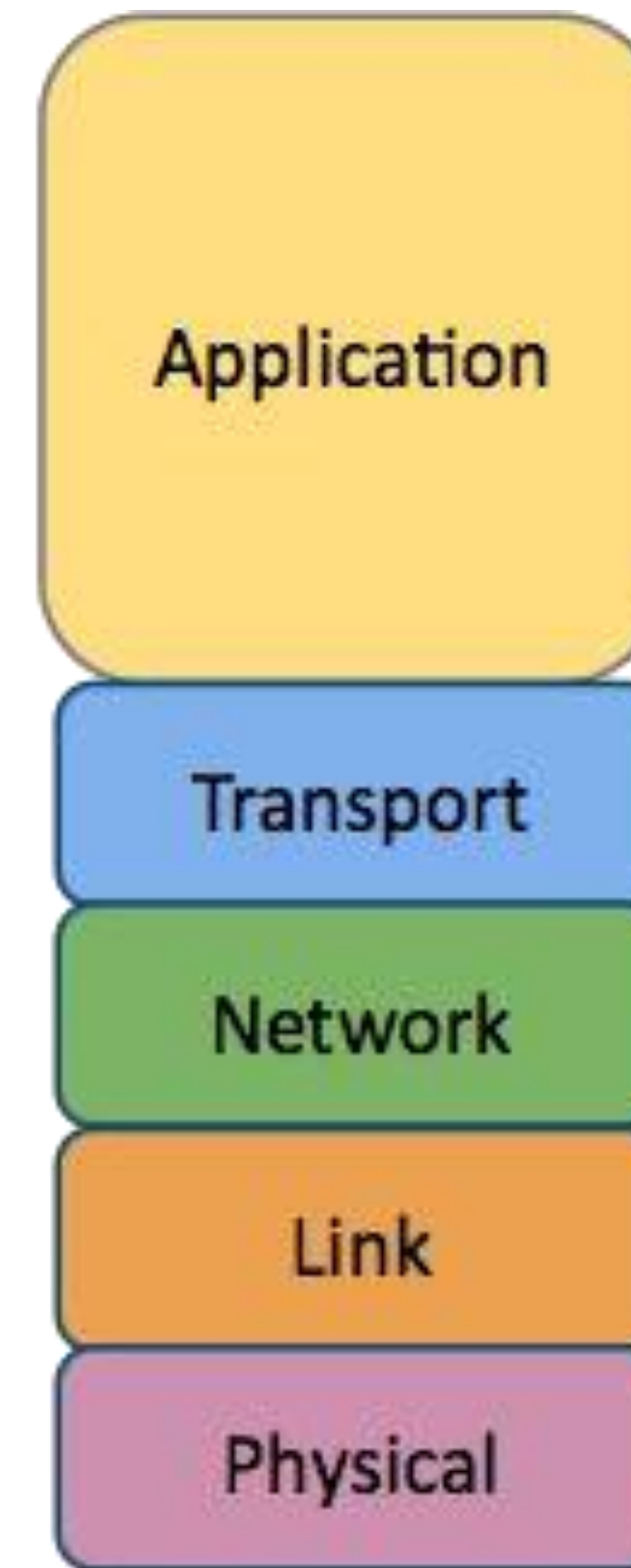


OSI Model x TCP/IP Model

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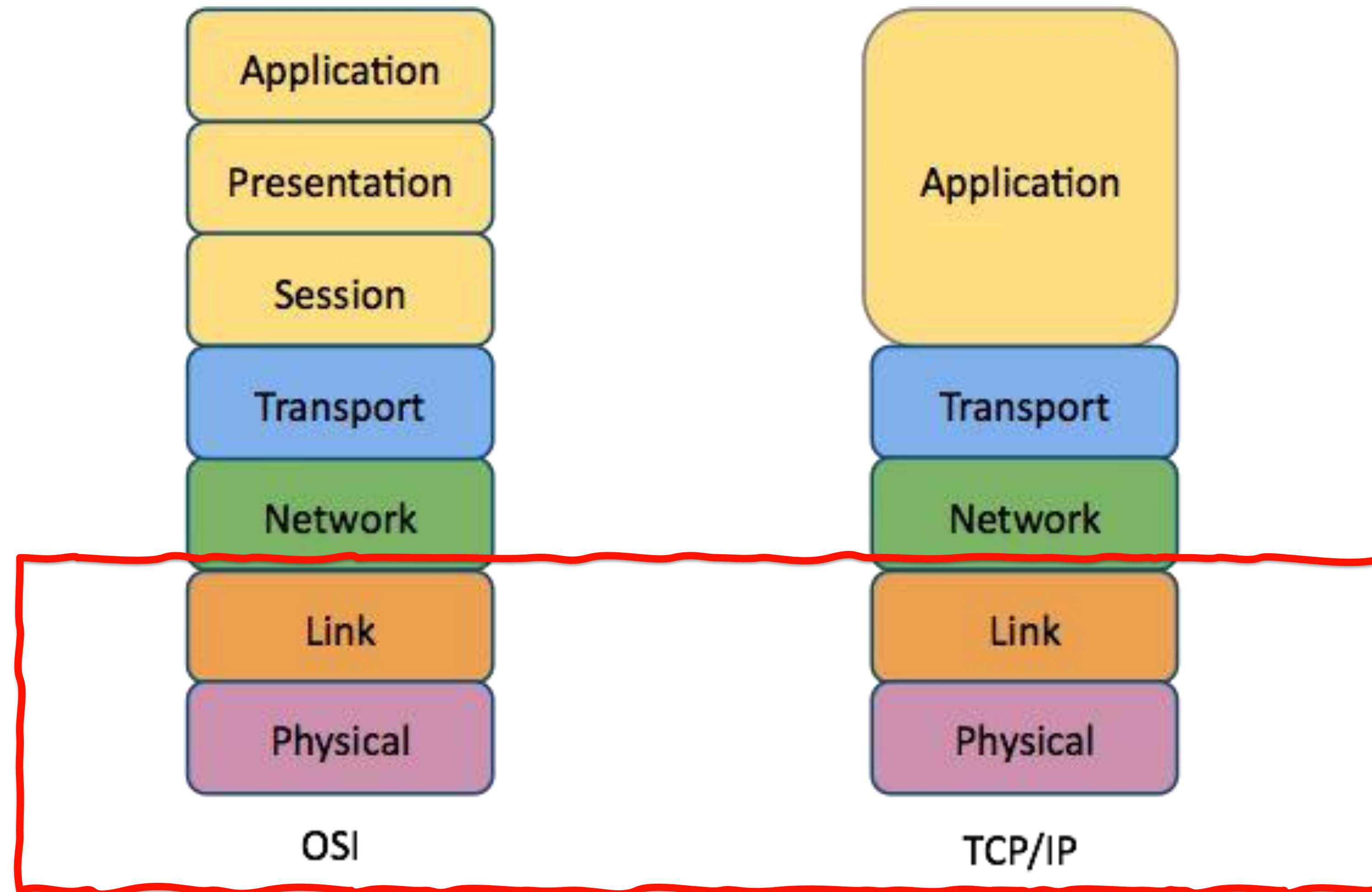
OSI



TCP/IP

OSI Model x TCP/IP Model

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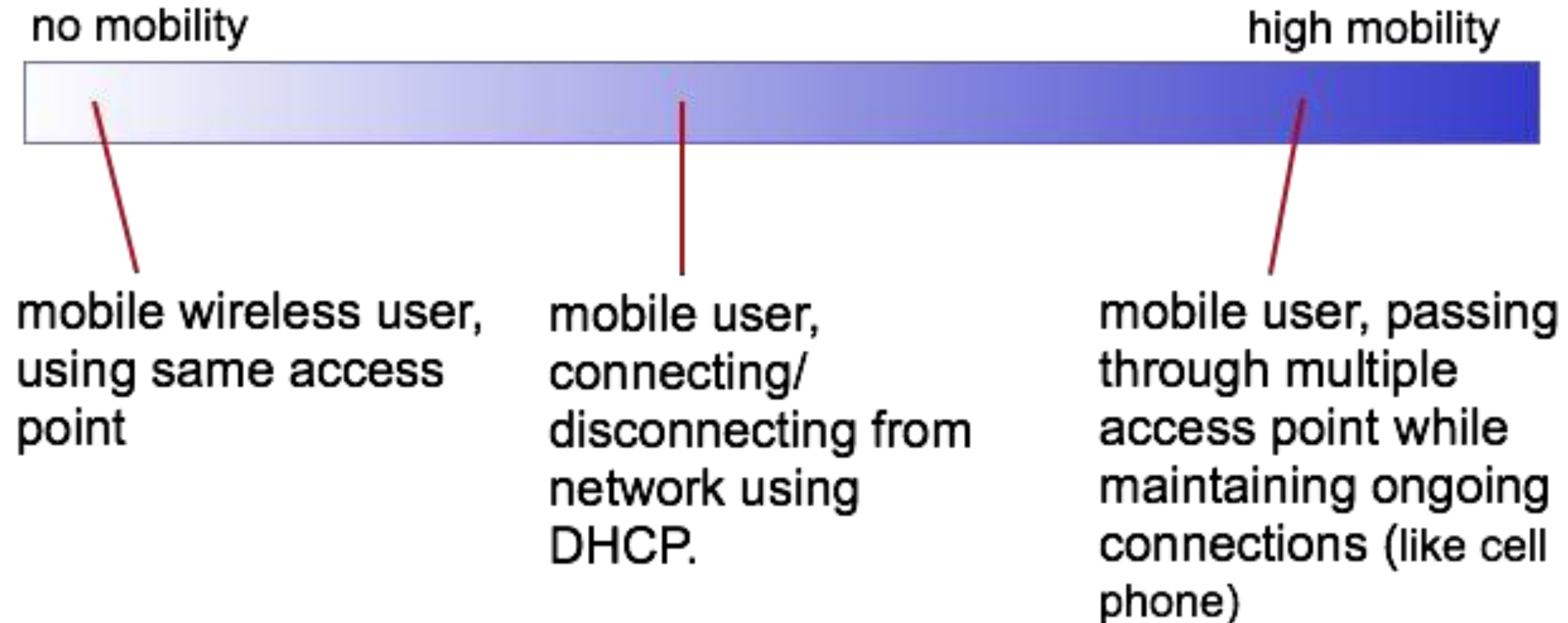
Wireless and Mobile Networks

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- Wireless X Mobile: two important (but different) challenges
 - wireless: communication over wireless link
 - mobility: handling the mobile user who changes point of attachment to network

Mobility

Spectrum of mobility, from the network perspective:



Wireless and mobility impact on higher layer protocols

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logically, impact should be minimal ...

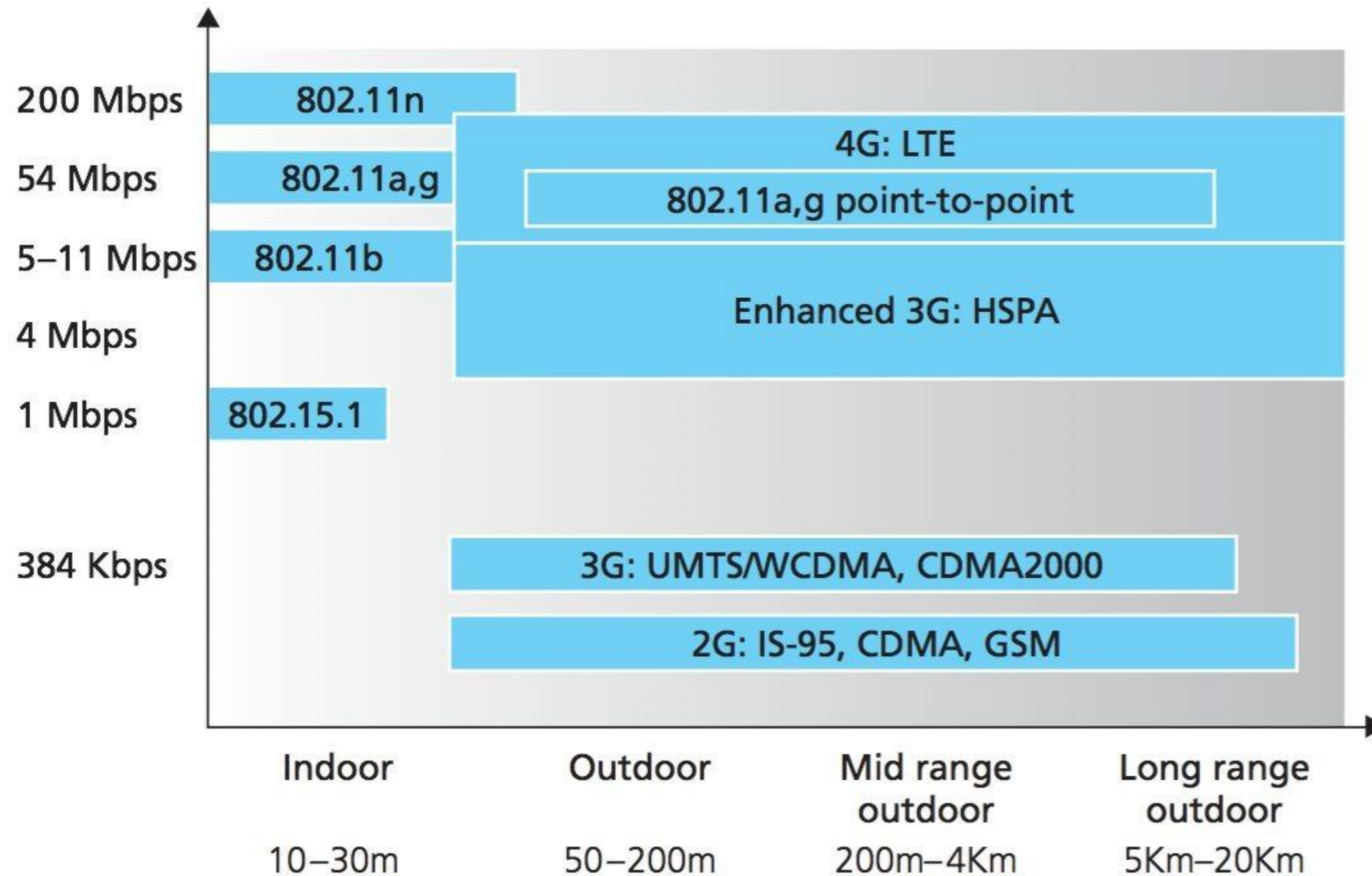
- best effort service model remains unchanged
- TCP and UDP can (and do) run over wireless, mobile

... but performance-wise:

- packet loss/delay due to bit-errors (discarded packets, delays for link-layer retransmissions), and handoff
- TCP interprets loss as congestion, will decrease congestion window unnecessarily
- delay impairments for real-time traffic
- limited bandwidth of wireless links

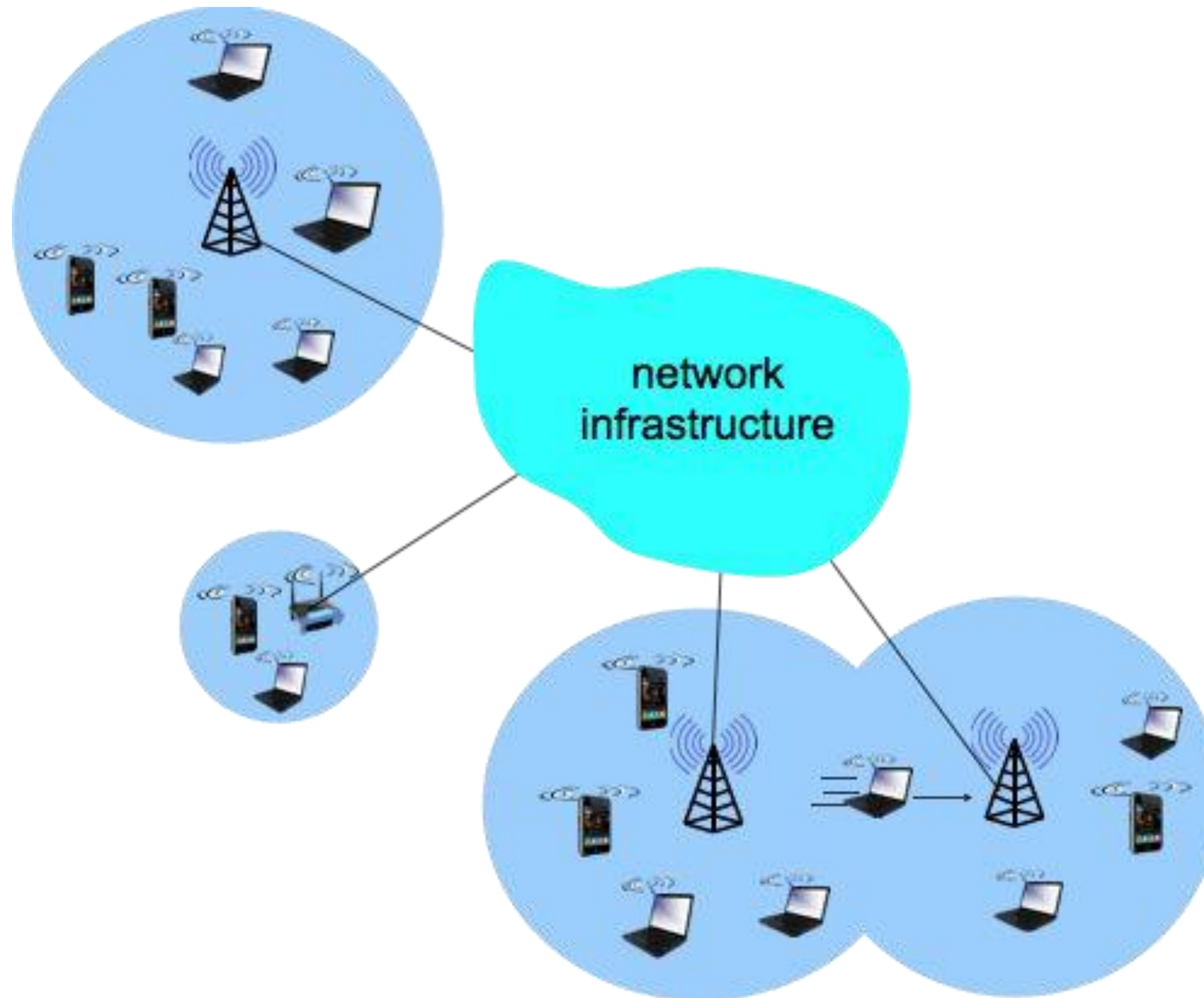
Key characteristics of popular wireless standards

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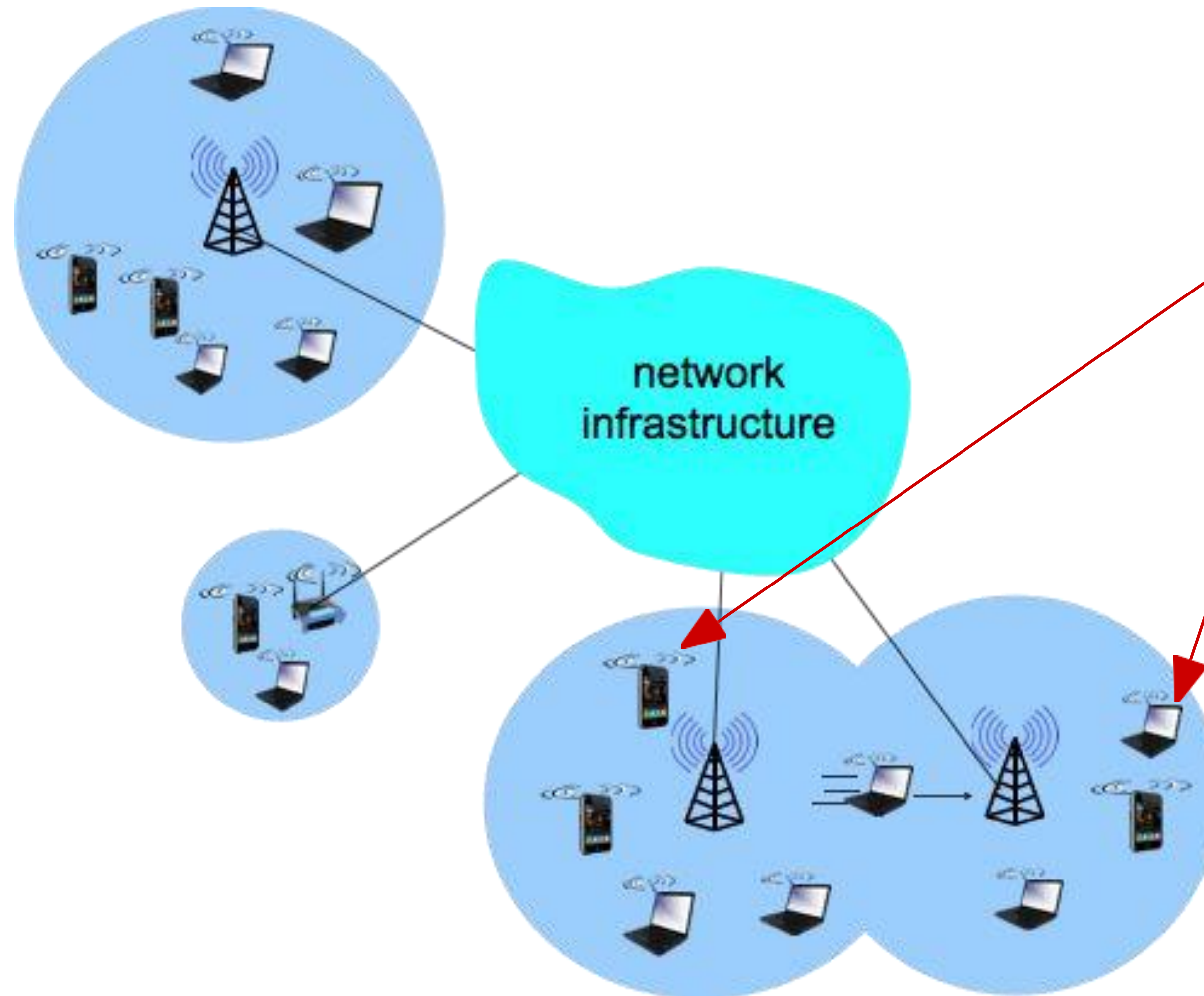
Elements of a wireless network

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Elements of a wireless network

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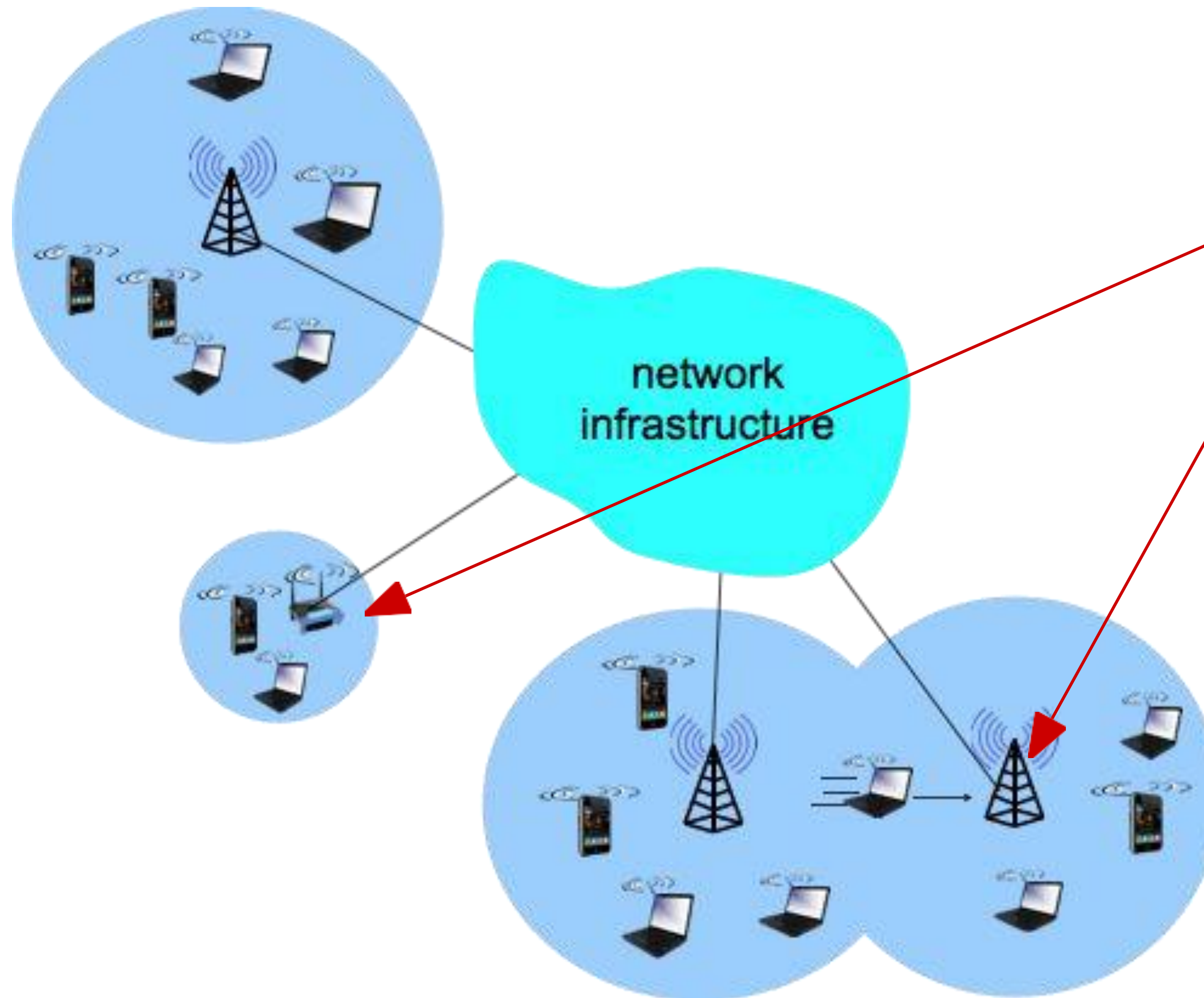


Wireless hosts:

- Laptops, smartphones
- Run applications
- May be stationary (non mobile) or mobile
 - Wireless does not always mean mobility.

Elements of a wireless network

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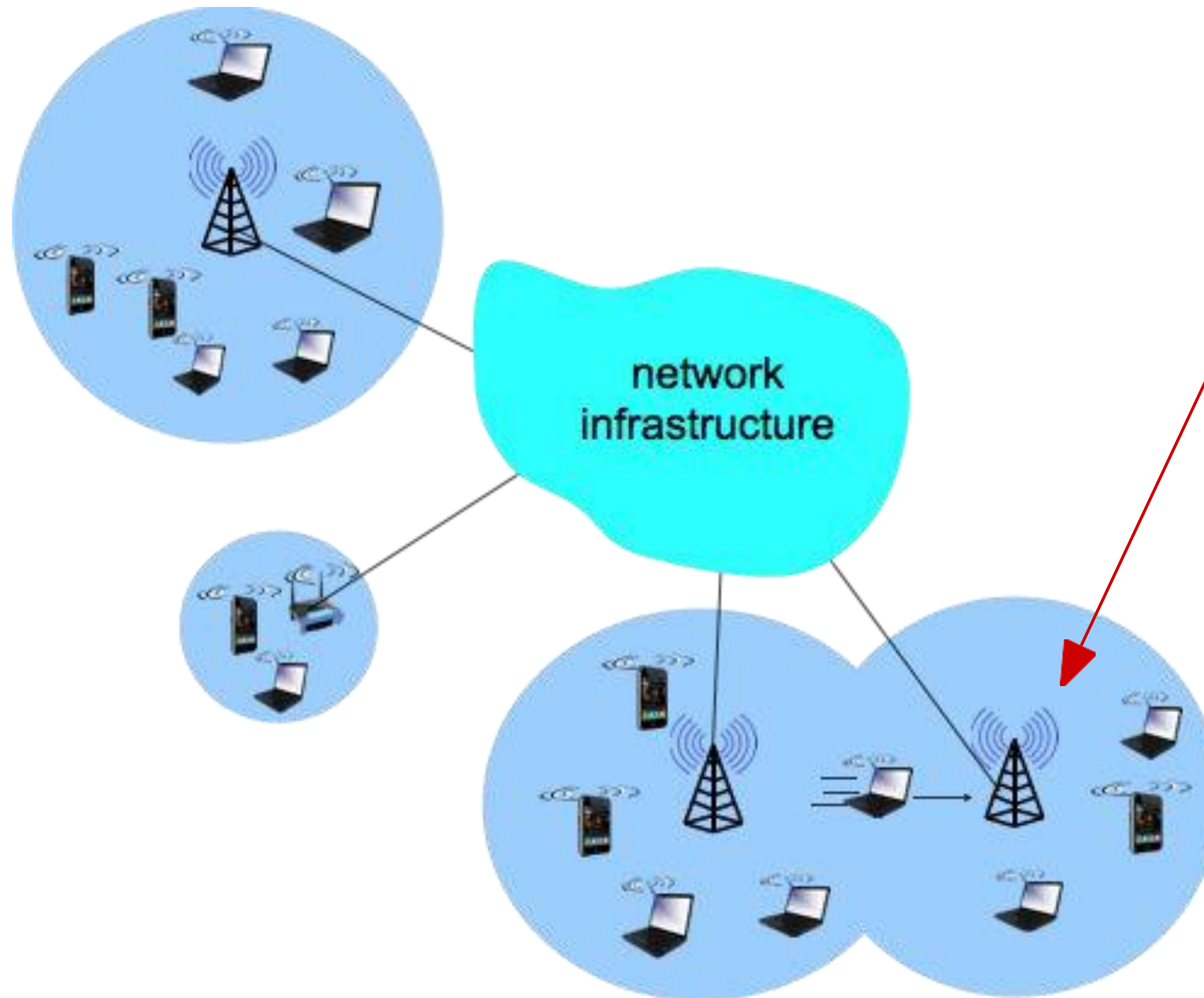


Base station

- typically connected to wired network
- responsible for sending packets between wired network and wireless host(s) in its "area"
 - e.g., cell towers, 802.11 access points

Elements of a wireless network

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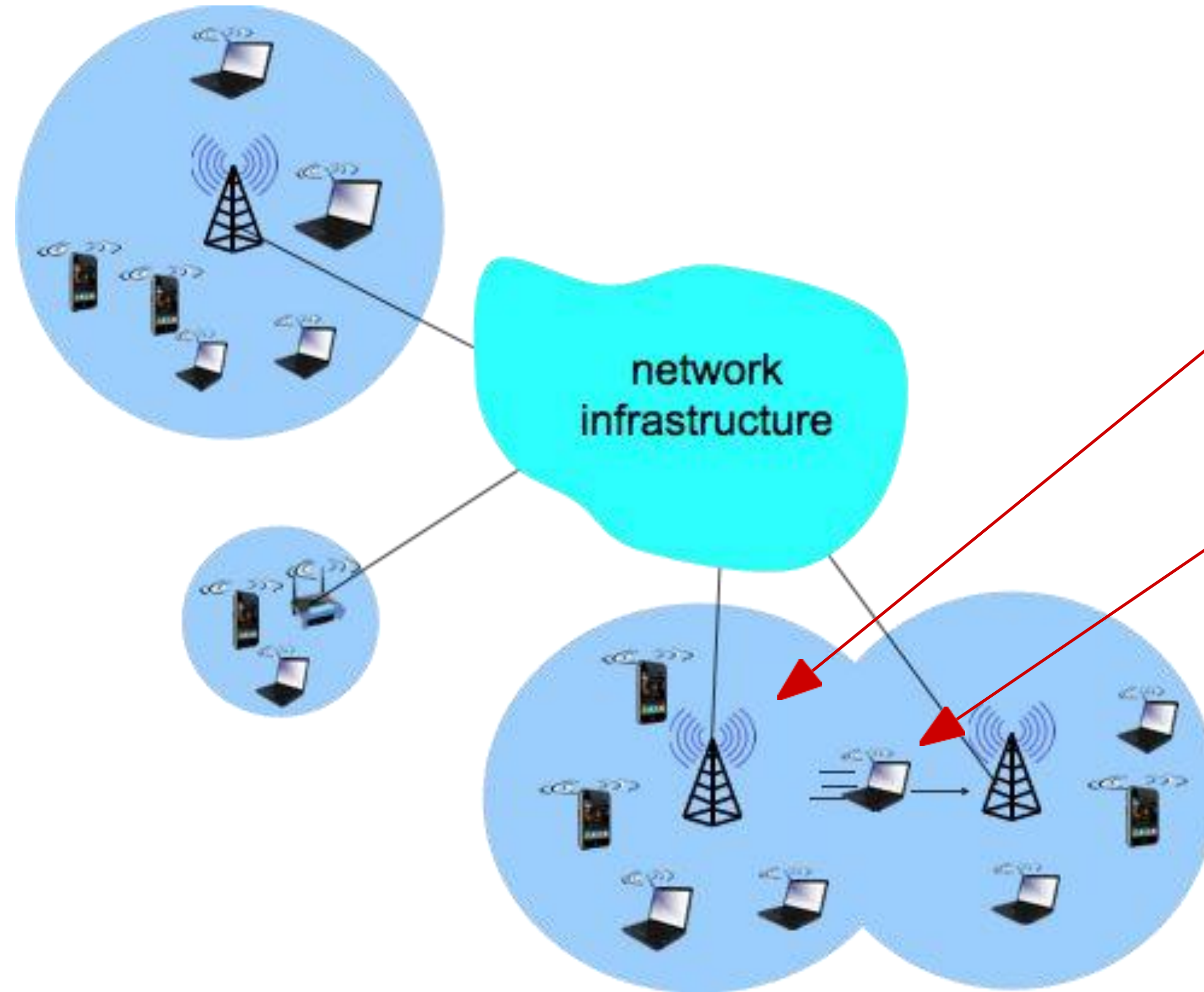


Wireless link

- typically used to connect mobile(s) to base station
- also used as backbone link
- multiple access protocol coordinates link access
- various data rates, transmission distance

Infrastructure x Ad hoc mode

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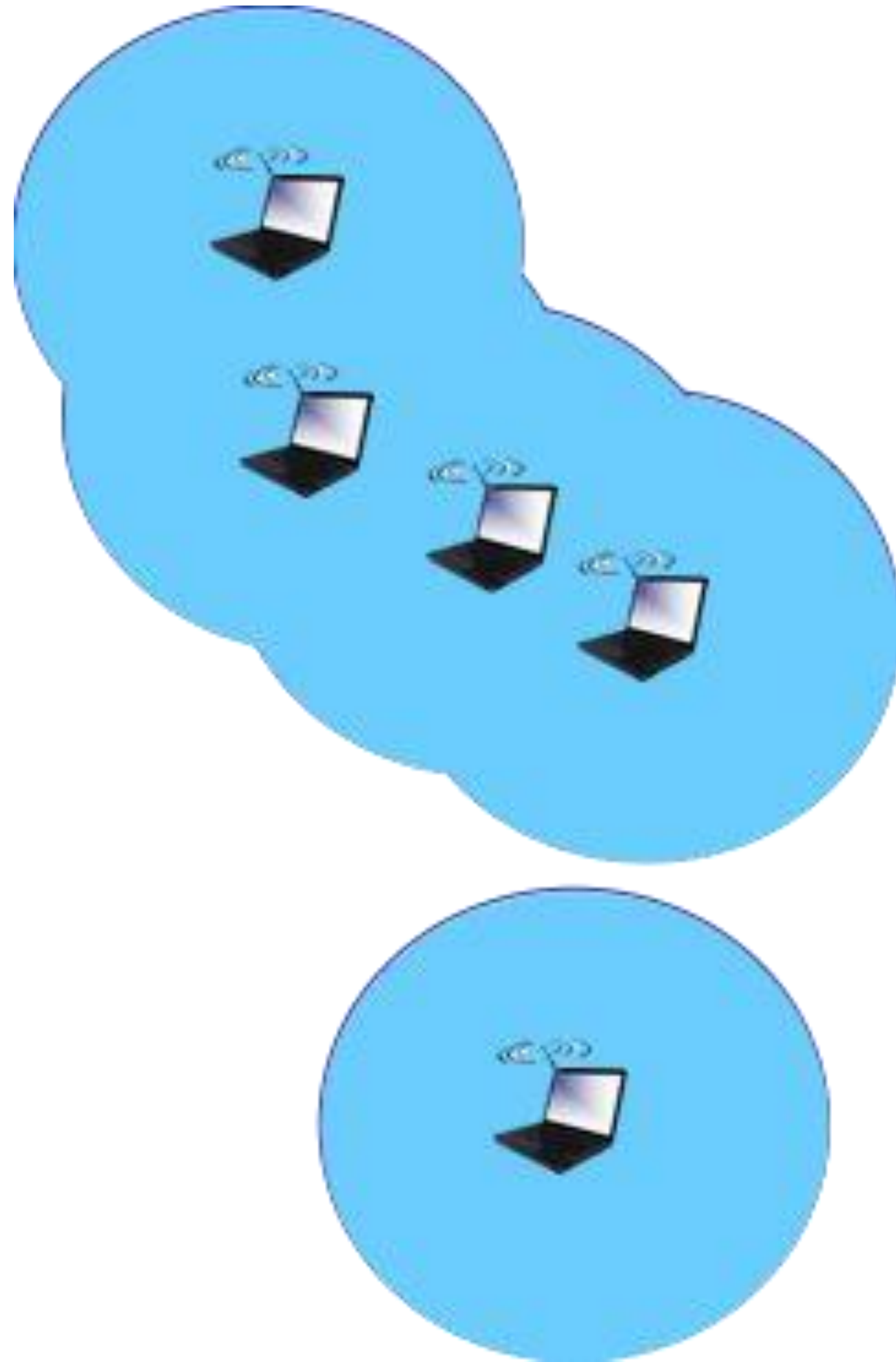


Infrastructure mode

- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

Infrastructure x Ad hoc mode

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Ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

Wireless network taxonomy

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At the highest level we can classify wireless networks according to two criteria:

- whether a packet in the wireless network crosses exactly one wireless hop or multiple wireless hops, and
- whether there is infrastructure such as a base station in the network

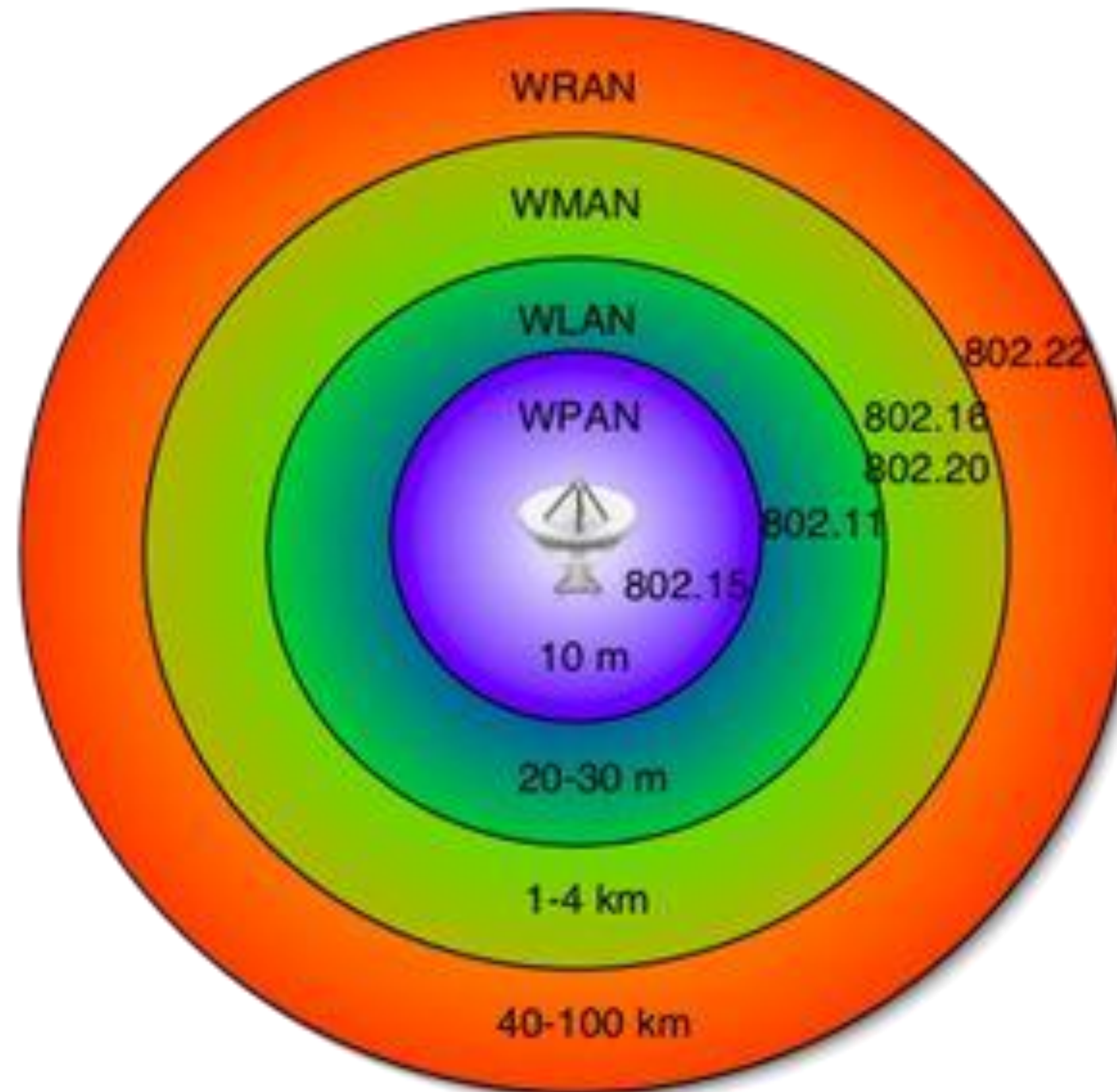
Wireless network taxonomy

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	Single hop	Multiple hops
Infrastructure	host connects to ^{base} station (WiFi, WiMAX, cellular) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet (mesh net)
No infrastructure	no base station, no connection to larger Internet (ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach a given wireless node

Wireless networks classification

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Source: <https://www.cse.wustl.edu/~jain/cse574-06/ftp/wimax/index.html>

Wireless link characteristics

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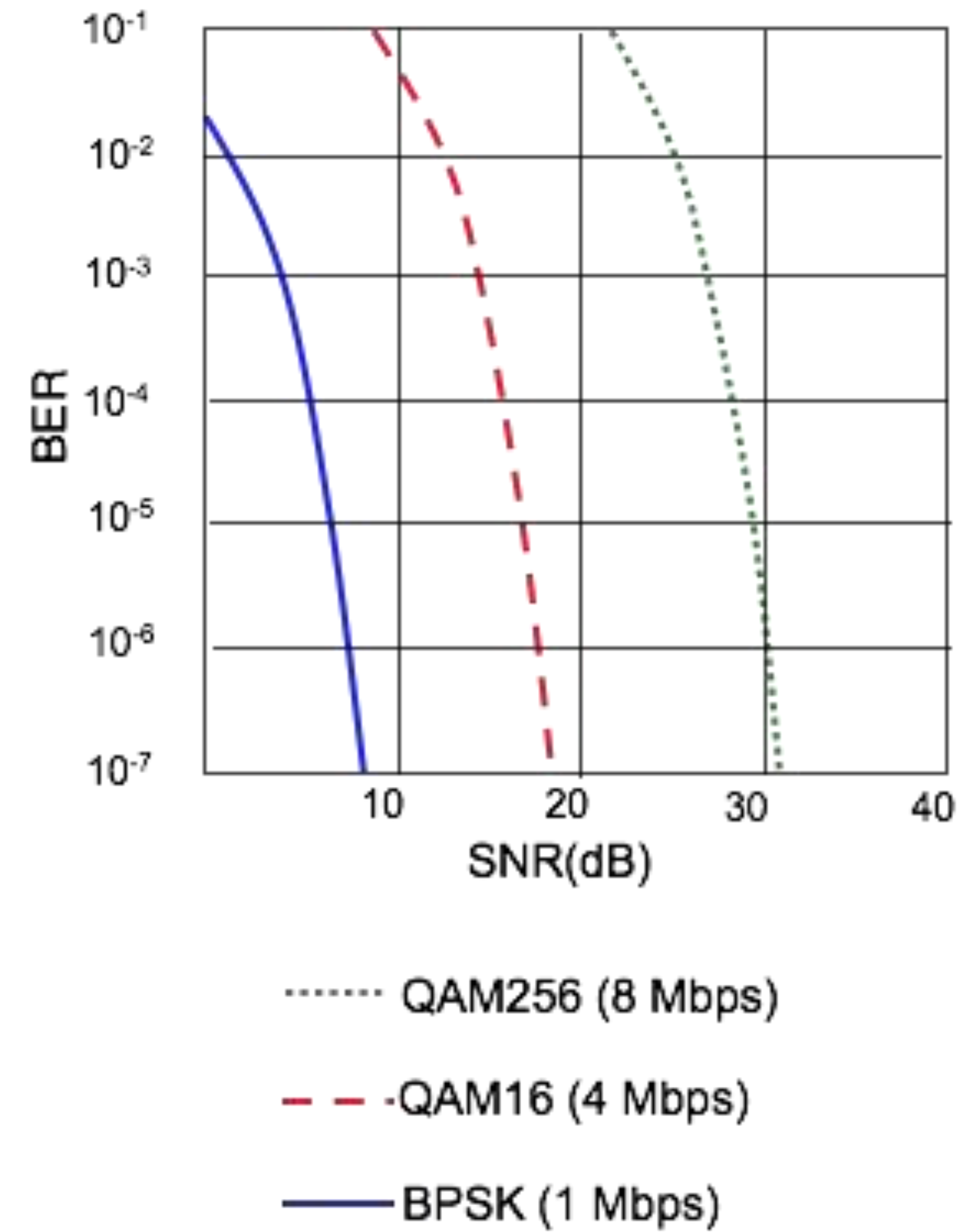
important differences from wired link

- **decreased signal strength:** radio signal attenuates as it propagates through matter (path loss)
- **interference from other sources:** standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- **multipath propagation:** radio signal reflects off objects ground, arriving at destination at slightly different times make communication across (even a point to point) wireless link much more “difficult”

Wireless link characteristic

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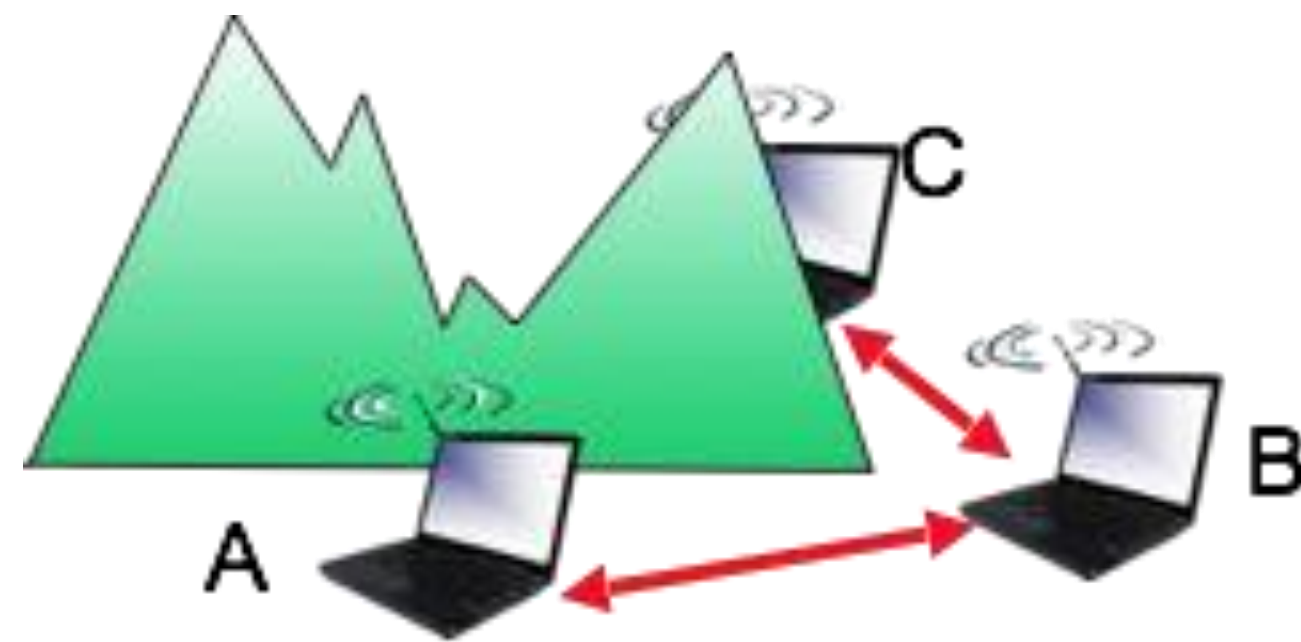
- SNR: signal-to-noise ratio
 - larger SNR – easier to extract signal from noise (a “good thing”)
- SNR versus BER (bit error rate) tradeoffs
 - given physical layer: increase power -> increase SNR->decrease BER
 - given SNR: choose physical layer that meets BER requirement, giving highest throughput
- SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)



Wireless network characteristics

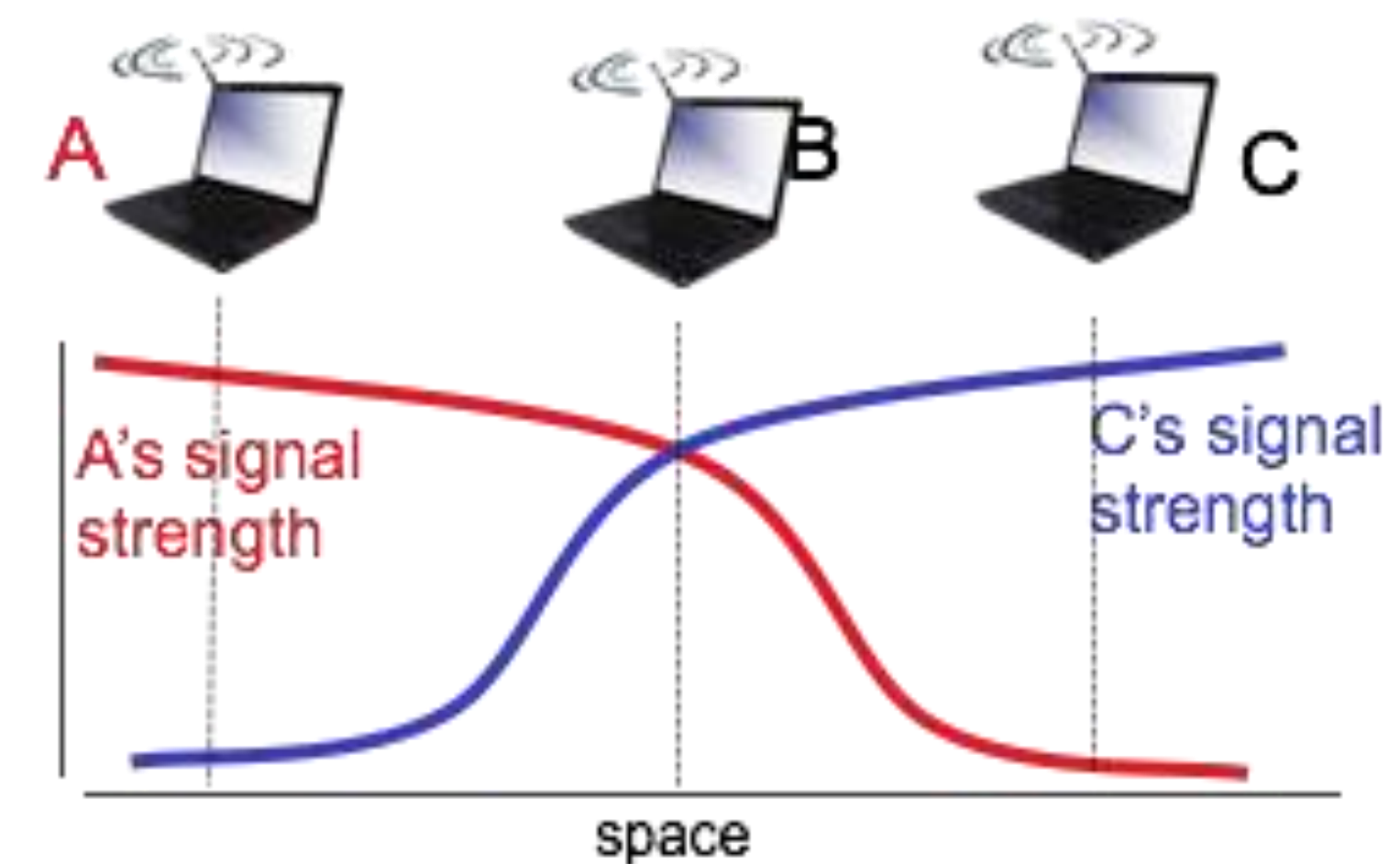
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Multiple wireless senders and receivers create additional problems (beyond multiple access):



Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B

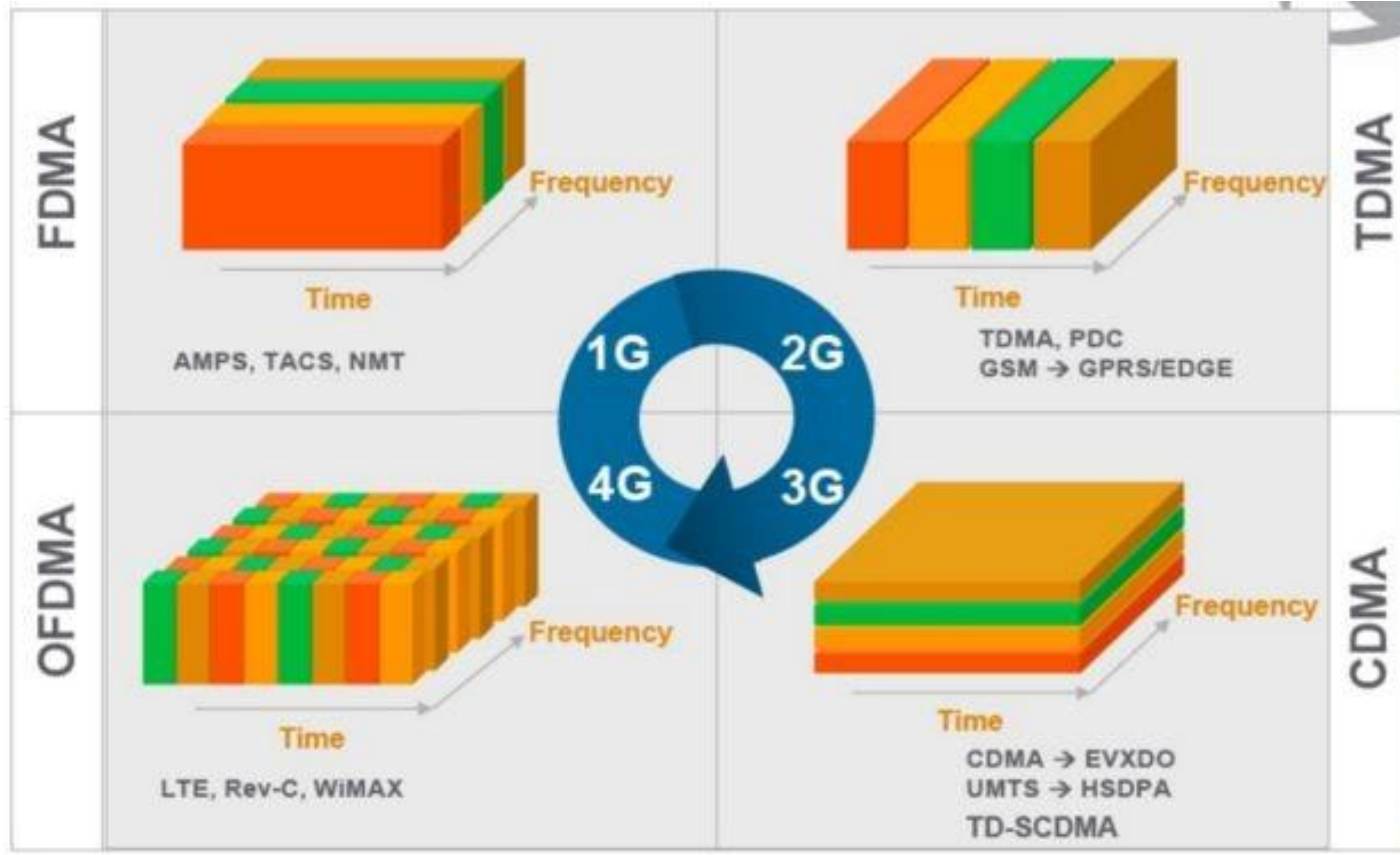


Signal attenuation:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

Multiple Access Methods

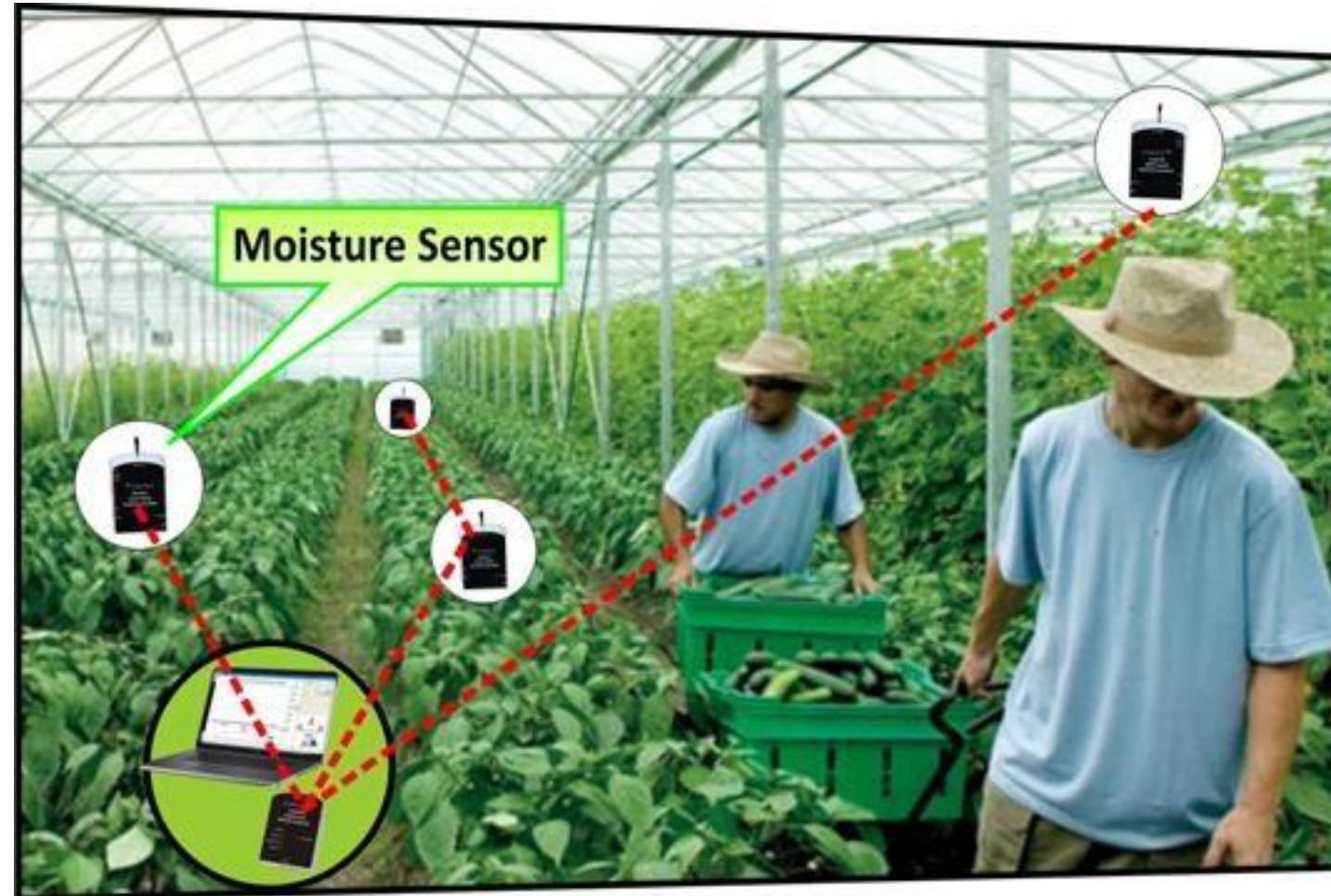
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Source: Nortel

Wireless sensor networks: applications

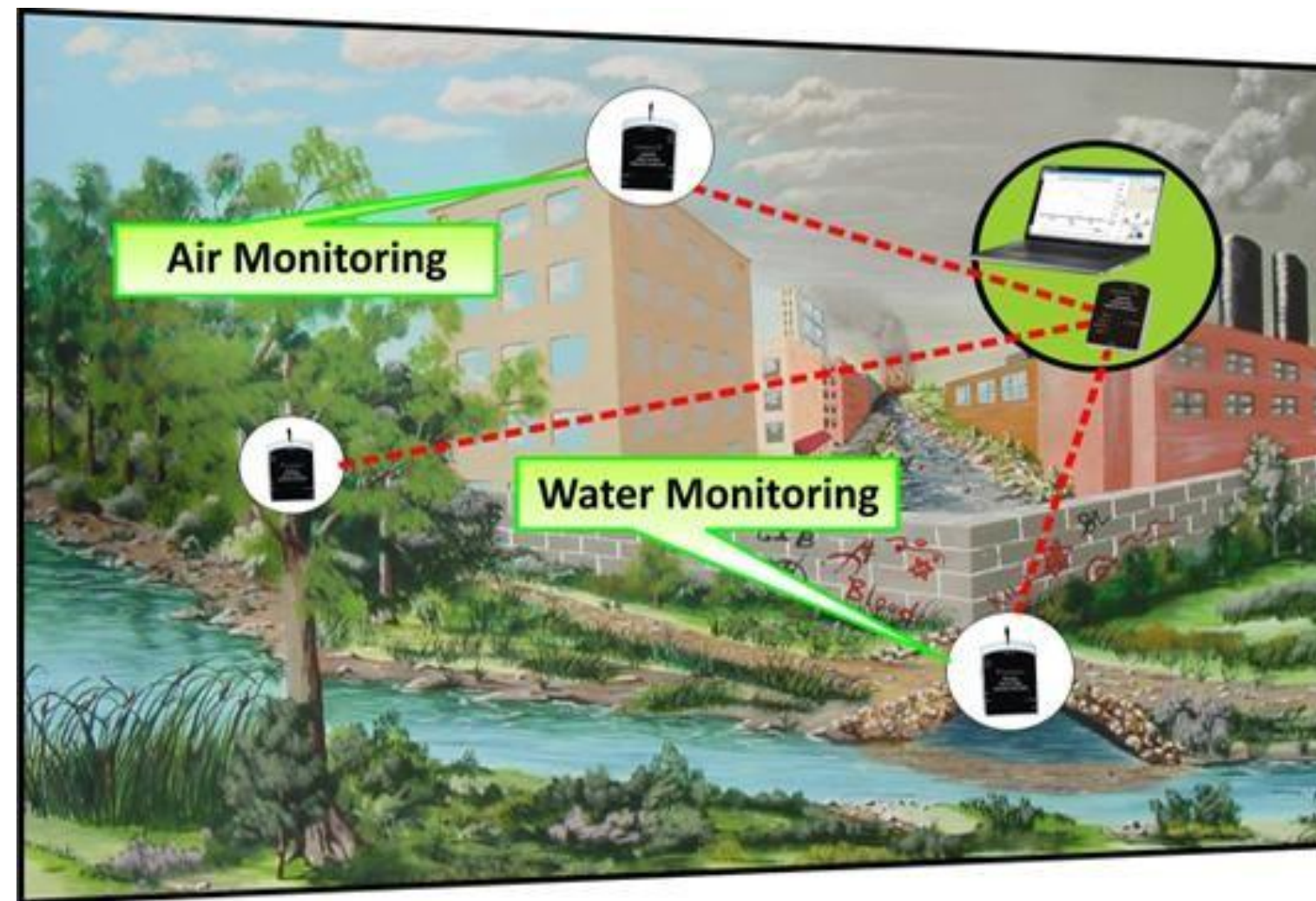
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Agriculture

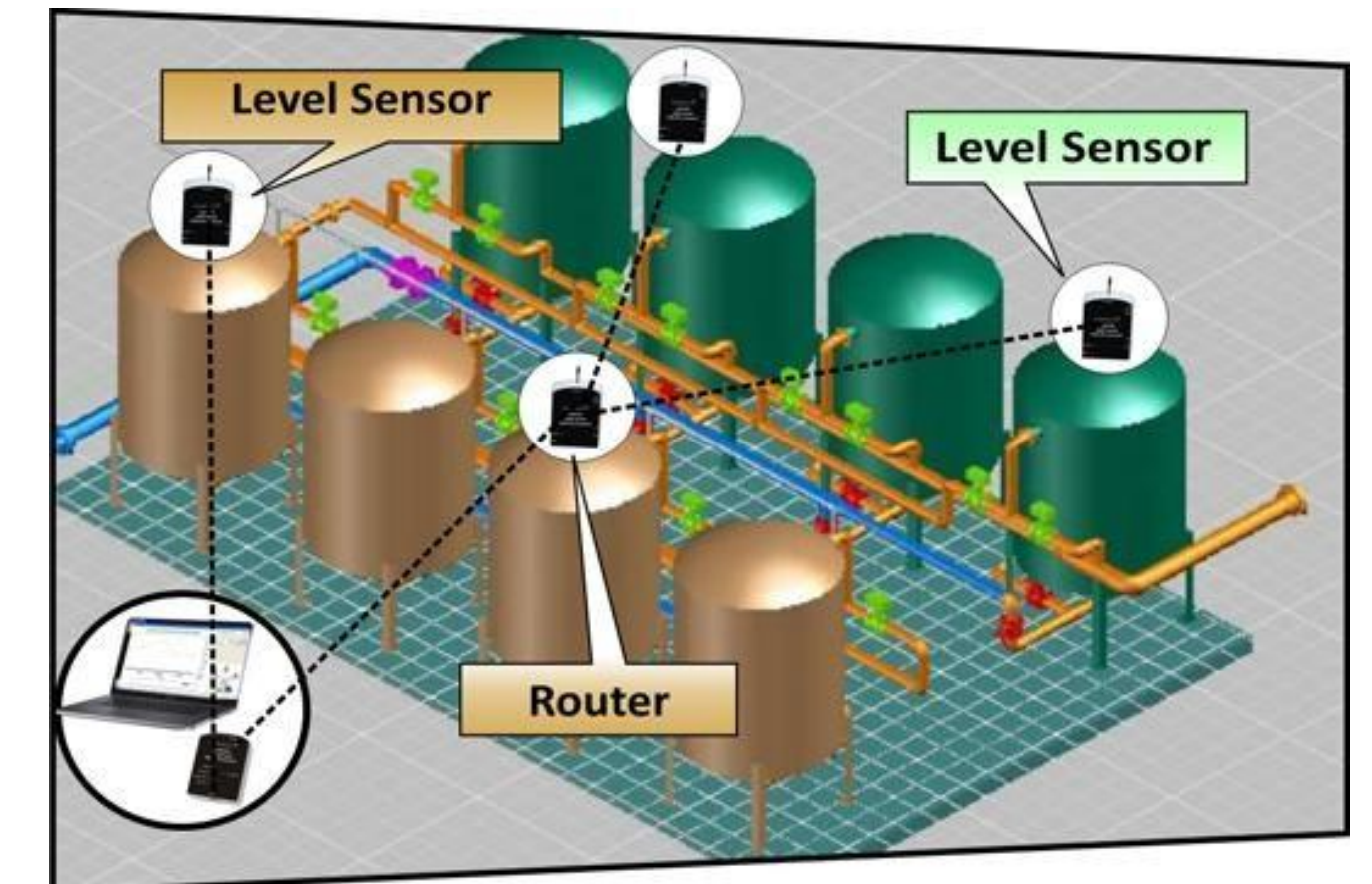


Structures



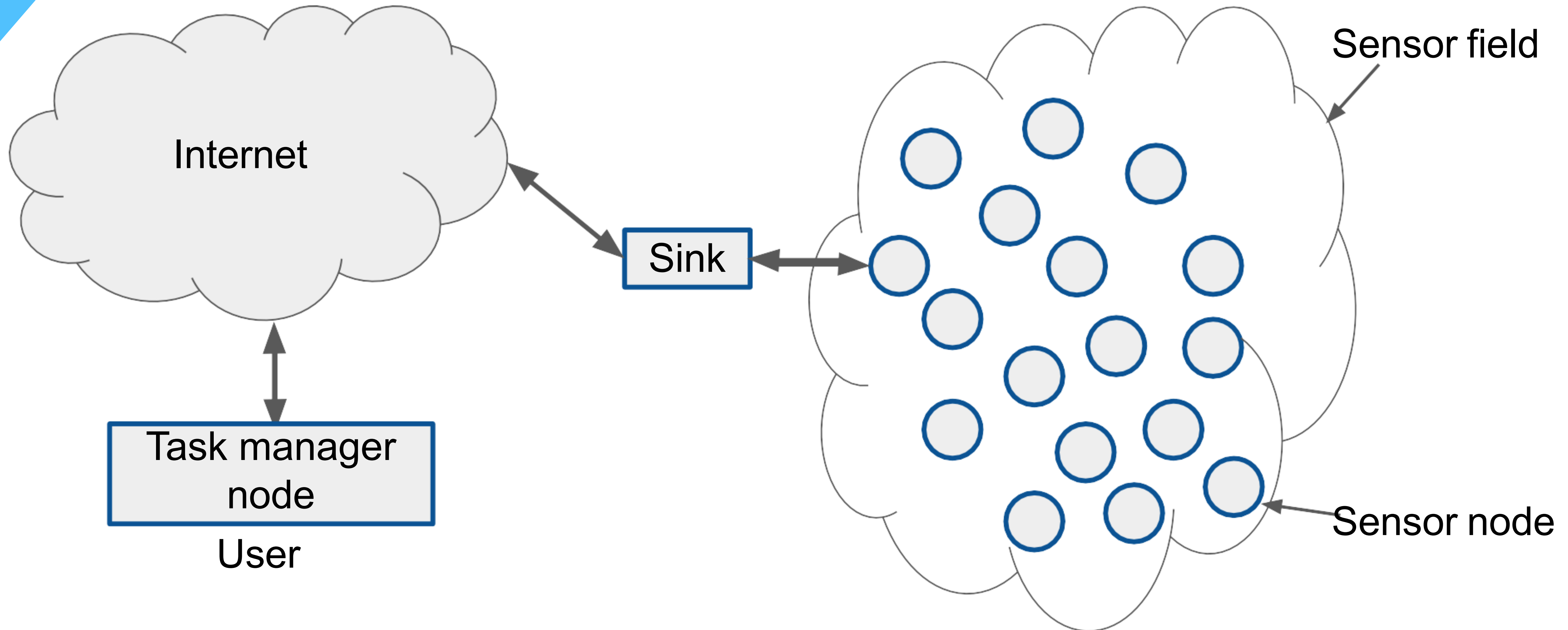
Environment

Industry



Wireless sensor networks: communication architecture

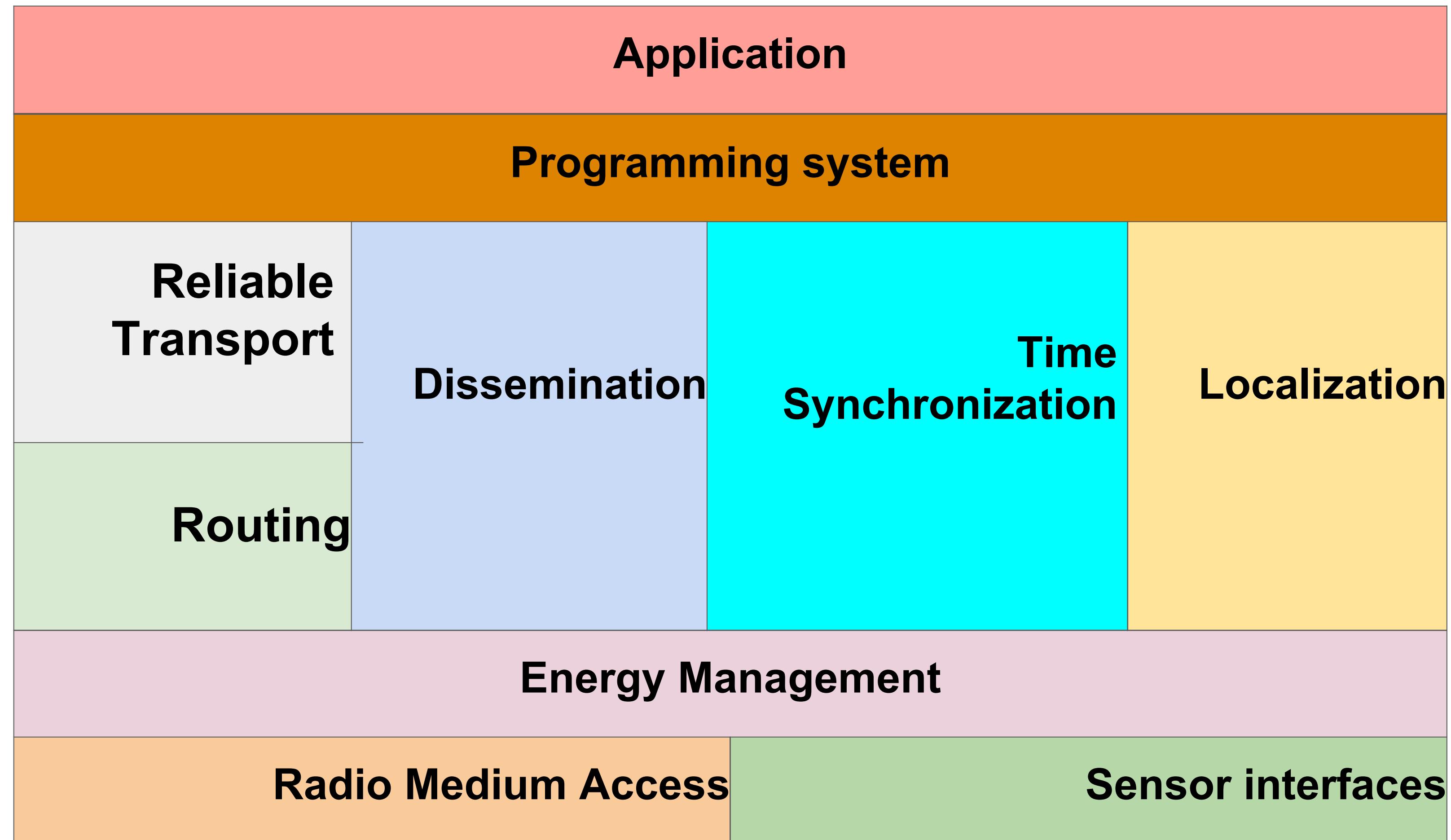
27



Source: I. F. Akyildiz *et al*, Wireless sensor networks: a survey. Computer Networks, 38:4, pp. 393-422, 2002.

Wireless sensor networks: software architecture

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Source: M. A. Vieira, Redes de Sensores Sem Fio Terrestres. Escola de Verão, DCC/UFMG, 2013.

Communication patterns

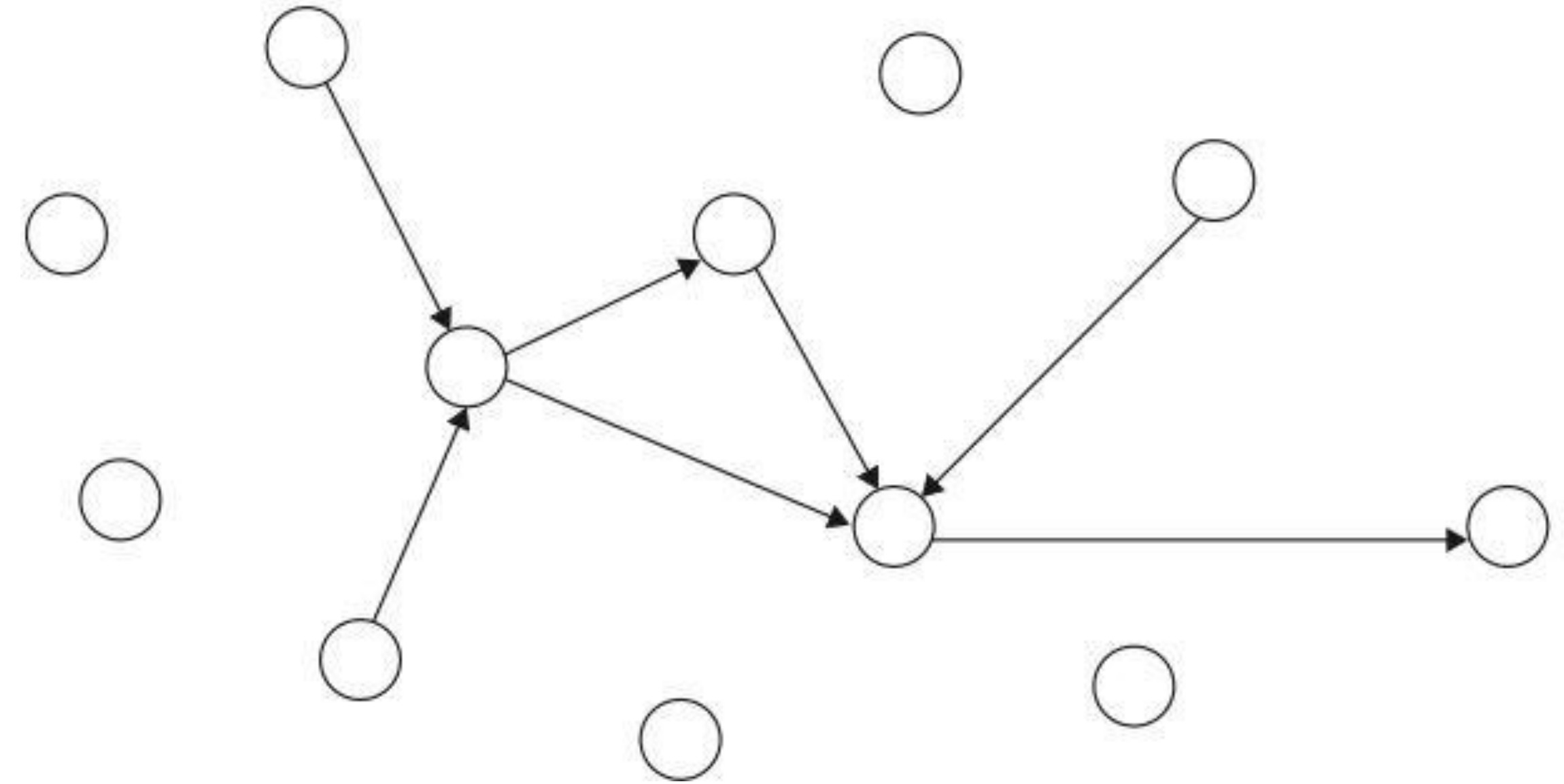
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- 3 categories: one-to-one, one-to-many, many-to-one.
- Communication pattern will depend on: application, reliability and power consumption concerns.

Communication patterns: many-to-one

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- Data collection
- Tree rooted at the sink node
- Ex.: nodes send periodic status to sink node
- Data delivery is unreliable
 - Sink node is unable to acknowledge received messages

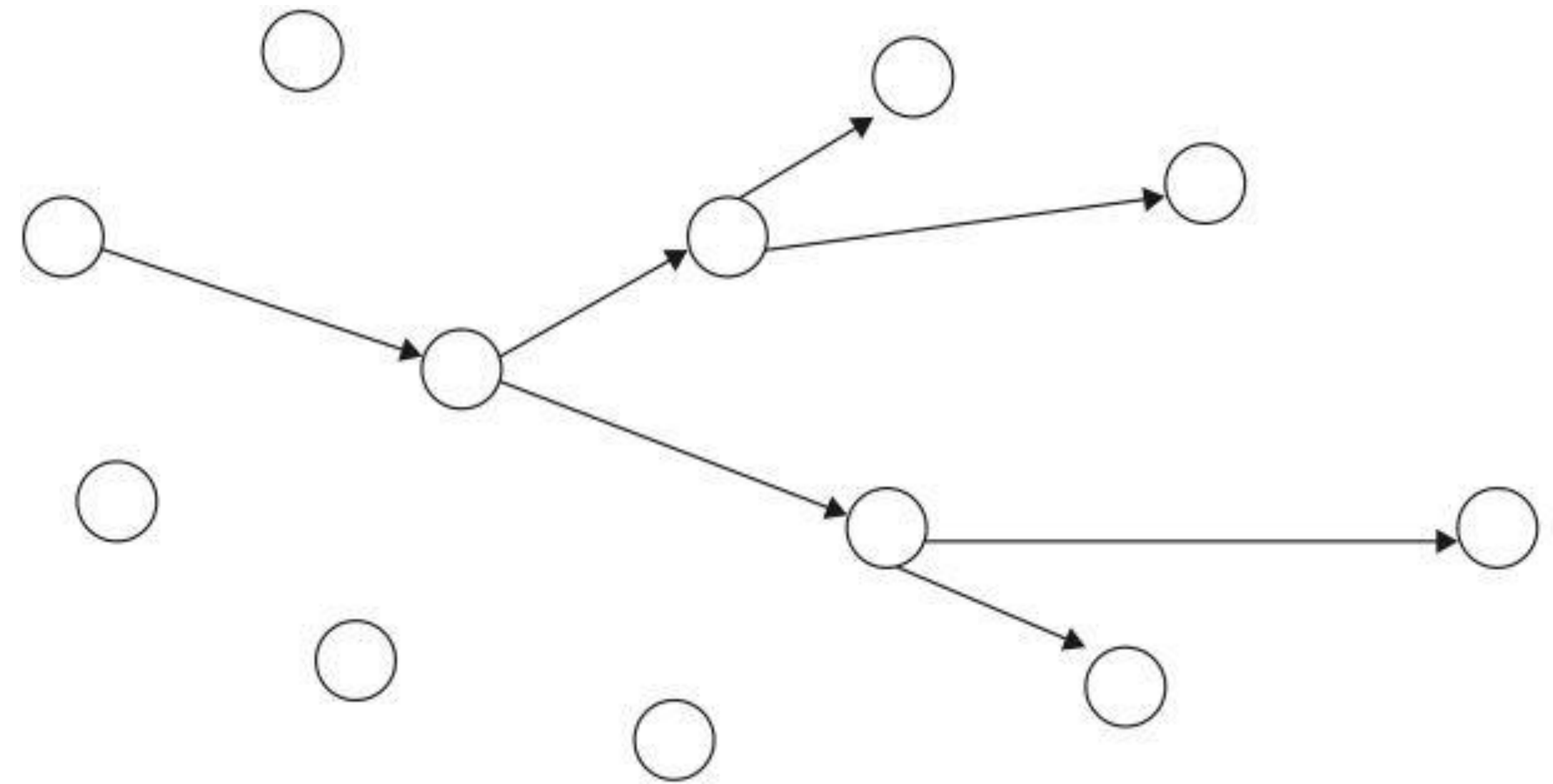


Source: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Communication patterns: one-to-many

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- Data dissemination
- Usually used to send commands or for devices reconfiguration
- Data delivery is unreliable
 - Nodes are unable to acknowledge received messages
- Cost in terms of number of messages can be high if flooding mechanisms are used.

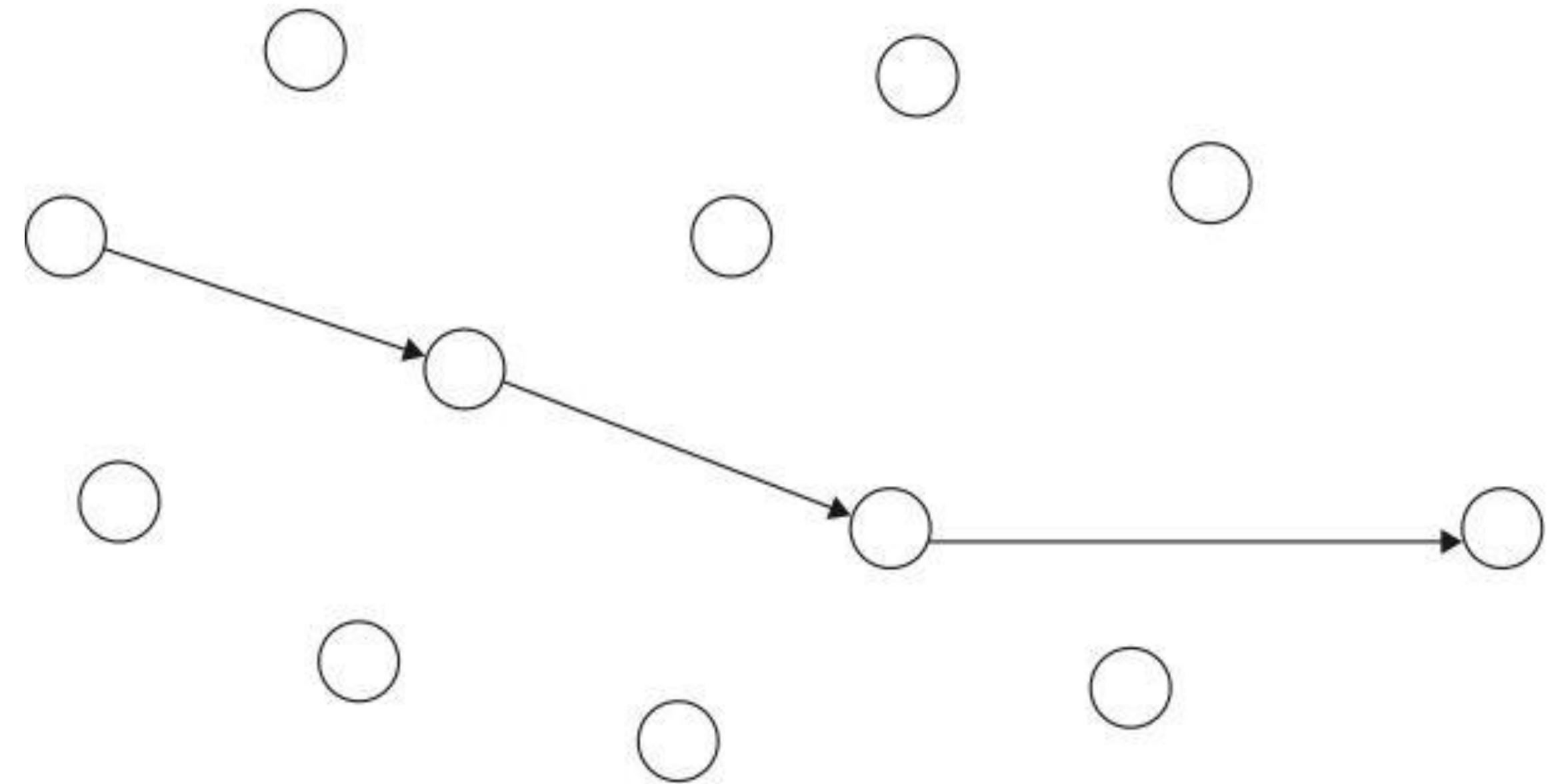


Source: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Communication patterns: one-to-one

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- Any two nodes can communicate
- More complex:
 - requires more resources
 - storage to routes to all reachable nodes
- More flexible for applications



Source: Jean-Philippe Vasseur and Adam Dunkels. 2010. Interconnecting Smart Objects with IP: The Next Internet. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Wireless sensor networks: Internet integration approaches

Front-end solution:

- external Internet hosts and sensor nodes never communicate directly with each other;
- WSN can implement its own set of protocols;
- external interactions managed by a centralized device (eg. base station);
- base station can store all the data streams coming from the WSN.

Wireless sensor networks: Internet integration approaches

Gateway solution:

- considers the existence of a device (e.g. base station) that acts as an application layer gateway;
- gateway translates the lower layer protocols from both networks (e.g. TCP/IP and proprietary);
- gateway routes the information from one point to another;

Wireless Sensor Networks: Internet Integration

35 Approaches

TCP/IP solution:

- sensor nodes implement the TCP/IP stack (or a compatible set of protocols such as 6LoWPAN in 802.15.4 networks);
- sensor nodes can be considered as full-fledged elements of the Internet;
- any Internet host can open a direct connection with sensor nodes, and vice versa;
- **this solution fully integrates the WSN with the IoT.**

Outline

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Introduction

5G and beyond

LiFi

Conclusion

5G: what is it?

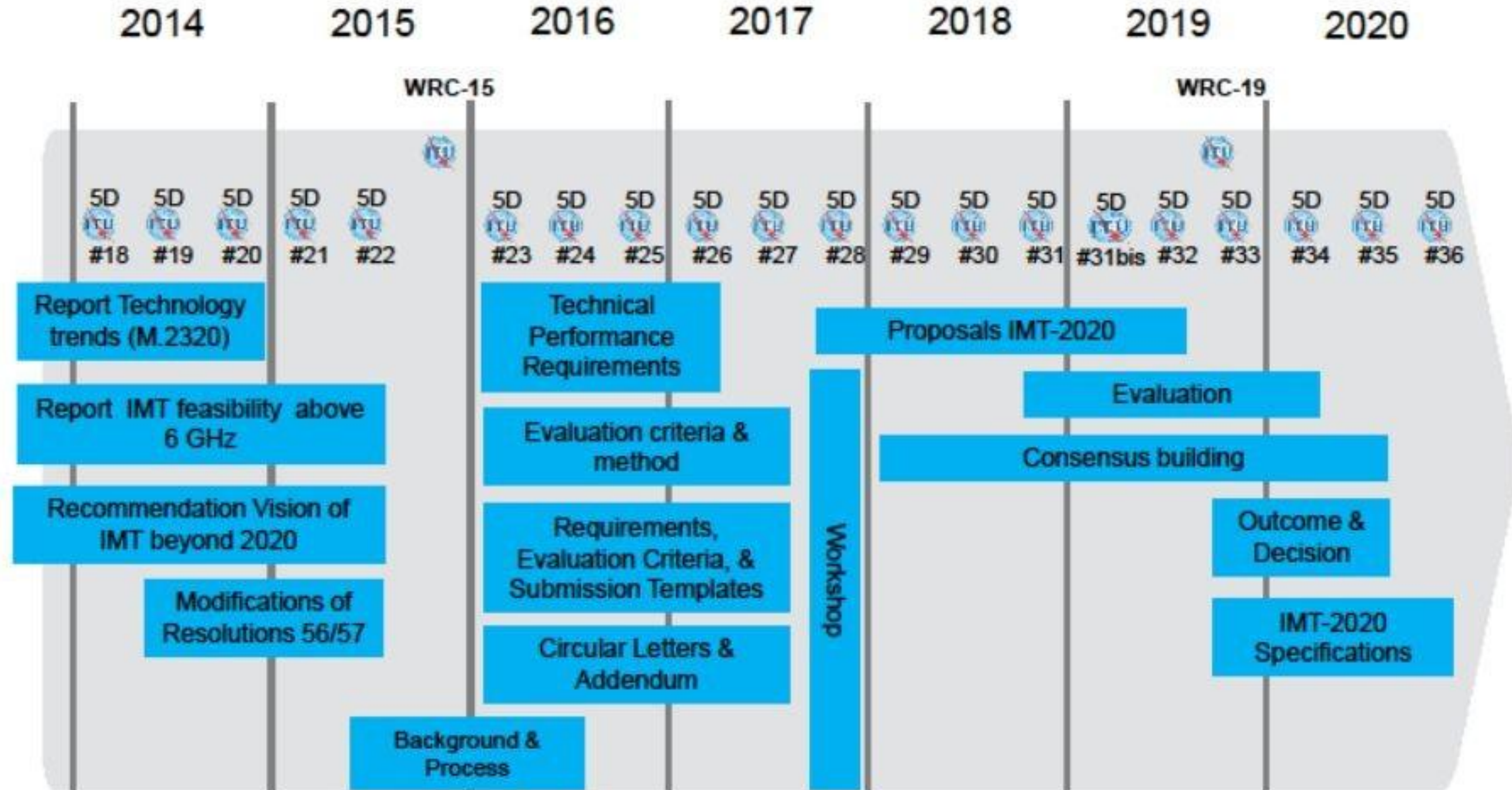
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- In early 2012 ITU-R initiates the development of "IMT for 2020 and beyond"
 - ITU-R = International Telecommunication Union - Radiocommunication Sector
 - IMT = International Mobile Telecommunications
- Focus of IMT-2020: to analyze how emerging 5G technologies will interact in future networks as a preliminary study into the networking innovations required to support the development of 5G system.

Source: www.itu.int

5G: what is it?

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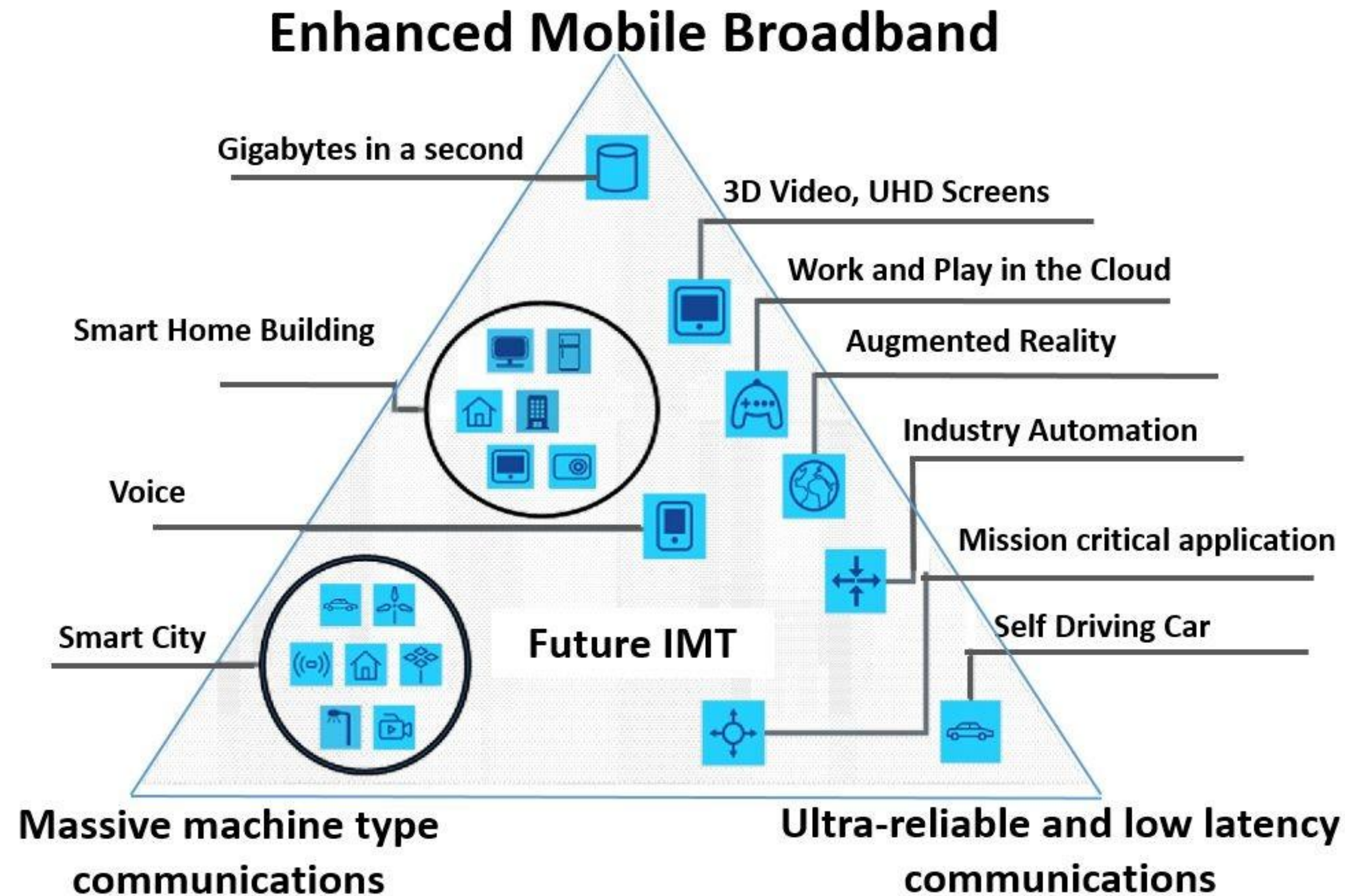


Source: ITU - Workplan, timeline, process and deliverables for the future development of IMT

5G: what is it?

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Source: ITU-R IMT 2020



5G: what is it?

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Key Performance Indicators	Value for 5G	Reference value for IMT-Advanced
User experienced data rate	100 Mbps - 1Gbps	10 Mbps
Peak data rate	20 Gbps	1 Gbps
Mobility	500 km/h	350 km/h
Latency	1 ms (radio interface)	10 ms (radio interface)
Connection density	10^6 per Km ²	10^5 per Km ²
Energy efficiency	100x greater than IMT-Advanced	
Spectrum efficiency	3x greater than IMT-Advanced	
Area Traffic Capacity	10 Mbps/Km ²	0.1 Mbps/Km ²

5G: enabling technologies

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**Millimeter
Waves**



Small Cell



**Massive
MIMO**



Beamforming



Full Duplex

Source: IEEE Spectrum ([video](#))

5G: enabling technologies

- Small cells!

- Factors that have contributed for the increase in the wireless capacity:
 - physical layer enhancements: 5x
 - spectrum: 25x
 - massive deployment of reduced cells: **1600x**

- **Source:** Mischa Dohler. The Future of Mobile Beyond 5G: Part

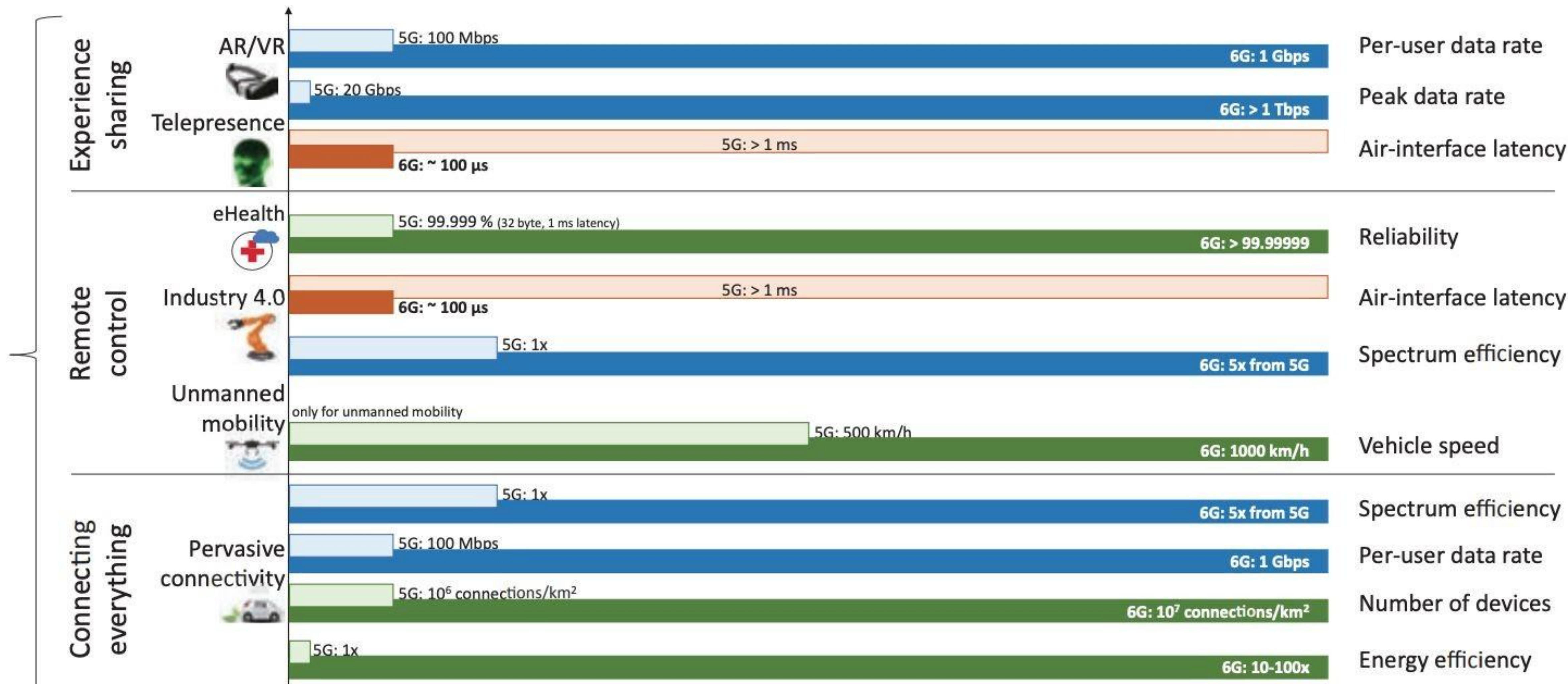
1. Available at:

<https://5g.ieee.org/publications/podcasts/future-mobile-1>

Source: Song Y, Choi W, Baek S (2017) Network switching strategy for energy conservation in heterogeneous networks. PLoS ONE 12(2): e0172318.

5G x 6G

6G use cases will merge these applications
e.g., unmanned mobility with VR/AR streaming



Source: M. Giordani, M. Polese, M. Mezzavilla, S. Rangan and M. Zorzi, "Toward 6G Networks: Use Cases and Technologies," in *IEEE Communications Magazine*, vol. 58, no. 3, pp. 55-61, March 2020

Outline

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IEEE 802.15.7 and LiFi

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LiFi: High-speed, bidirectional, networked and mobile wireless communications using light.

- **High speed:**
 - efficient use of optical bandwidth
 - data rates of at least 10 Mbps per access point
- **Bidirectional:**
 - light spectrum (190 nm – 10,000 nm wavelength)
 - Support full uplink and downlink communications without additional RF resources
- **Networked & Mobile:**
 - support roaming users and multiple users per access point (handover and multiple access)
 - user expectations of wireless technology

Secure & Safe:

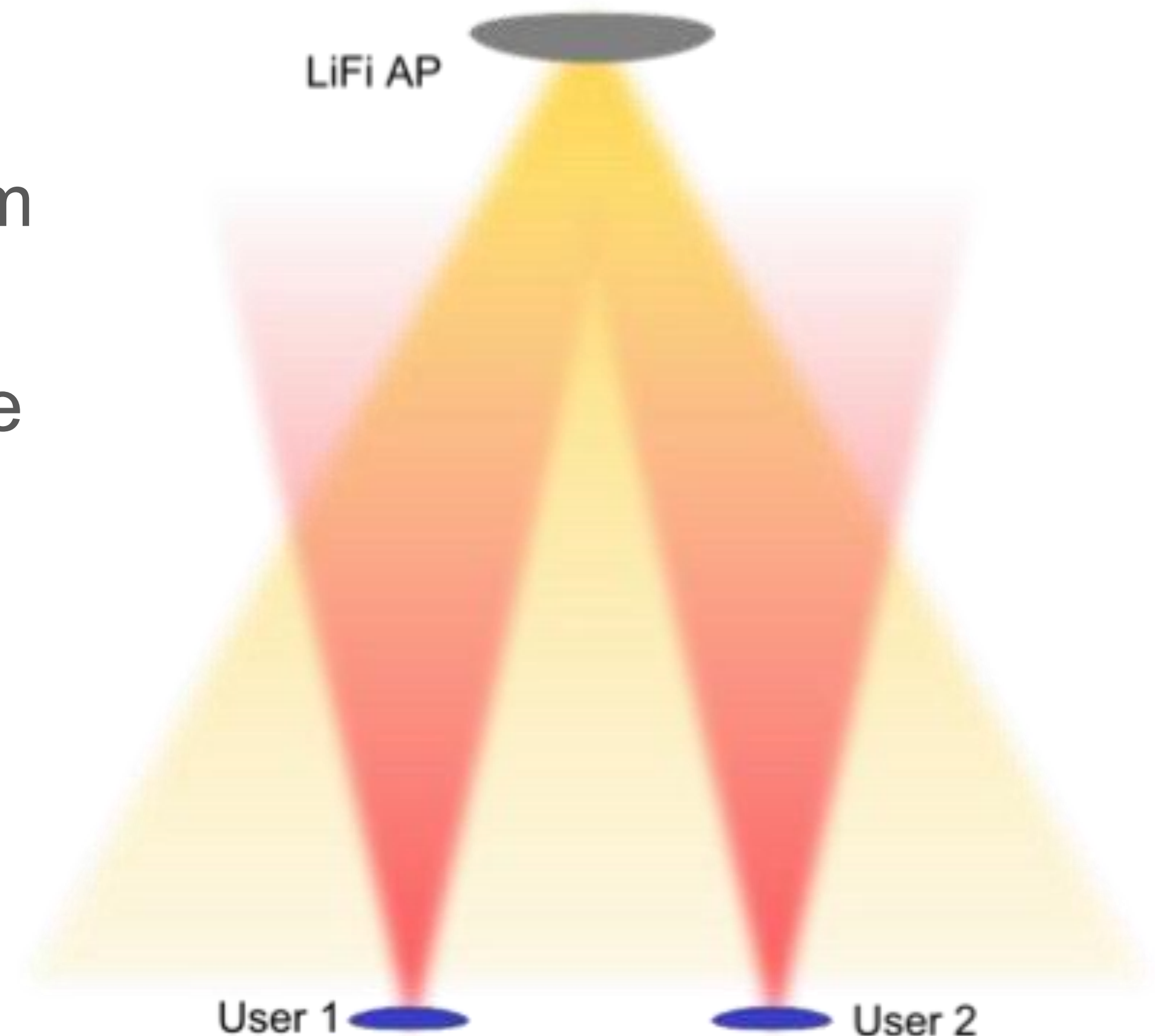
- localized => privacy, cybersecurity
- No electromagnetic interference (EMI) => good for EMI sensitive environments, e.g. planes.

LiFi: bidirectional and multiple access support

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Bidirectional:

- LiFi operates entirely in the light spectrum and can help alleviate the spectrum crunch;
- users request access to the AP which may be connected to the network
- users may see multiple APs;
- APs may see multiple users;
- the inherent directionality of light beams is used to define strict coverage regions.



LiFi: mobility and handover support

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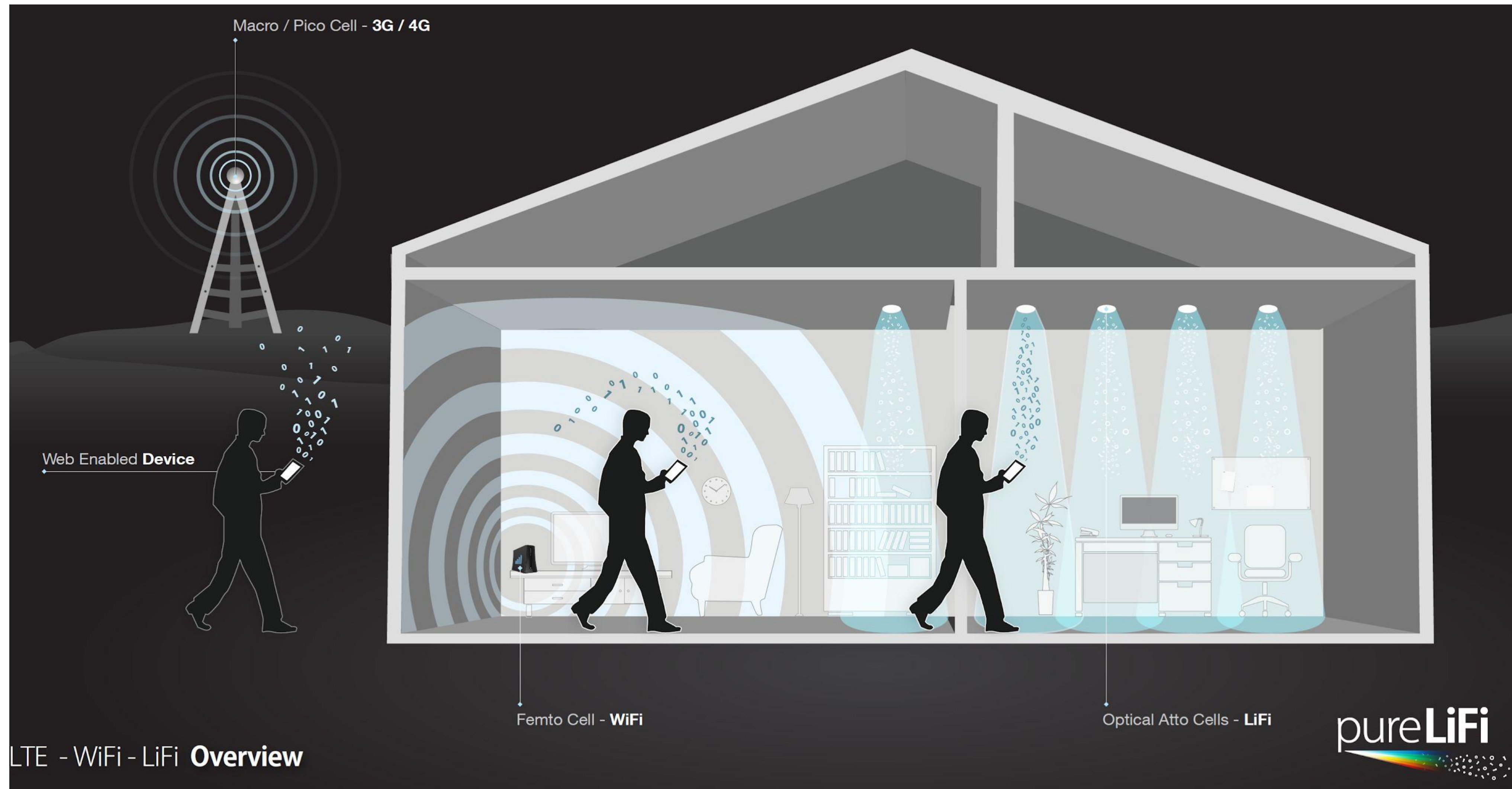
Networked & Mobile:

- users can move from one light source to another and retain their session;
- users do not need to be aligned with the light source and may connect to the strongest light source;
- the signal may be detected from reflections as well as direct line of sight.



LiFi: Complementary to existing radio frequency

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LiFi: Competitive Advantages

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- Security
 - If no photons escape, there is no external network intrusion.
- Localization
 - Strictly defined illumination area, sharp SINR cut-off, enables geo-fencing and asset tracking.
- Safety
 - No RF radiation alleviating health and electromagnetic interference (EMI) concerns.
- Data Density
 - Smaller LiFi cells allow significantly greater spatial reuse of bandwidth.

LiFi: use-cases

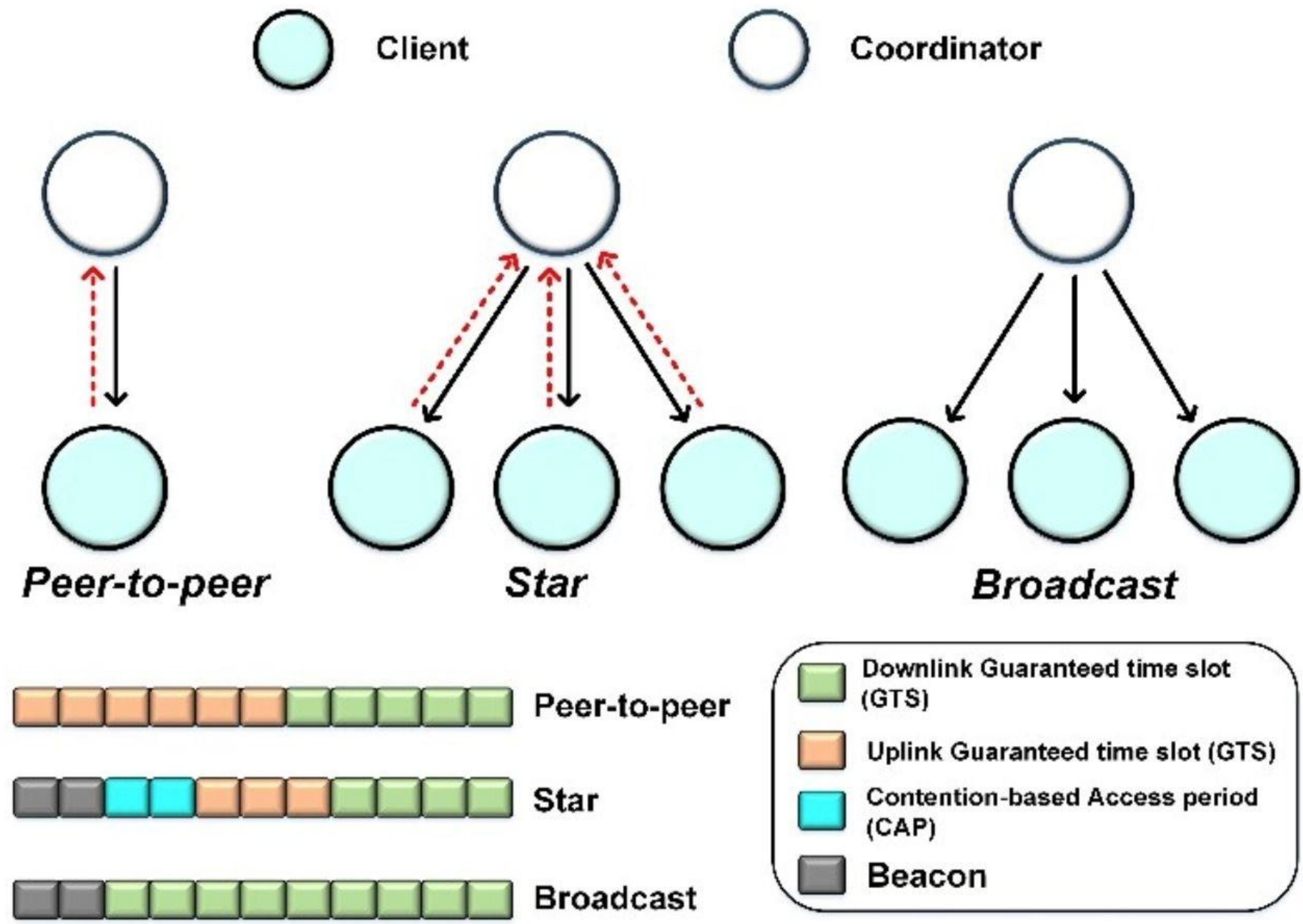
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- Security Conference rooms
- Airplanes and Airports
- EMI sensitive environments
 - hospitals, petrochemical plants, nuclear facilities, etc.



IEEE 802.15.7 - Network topologies

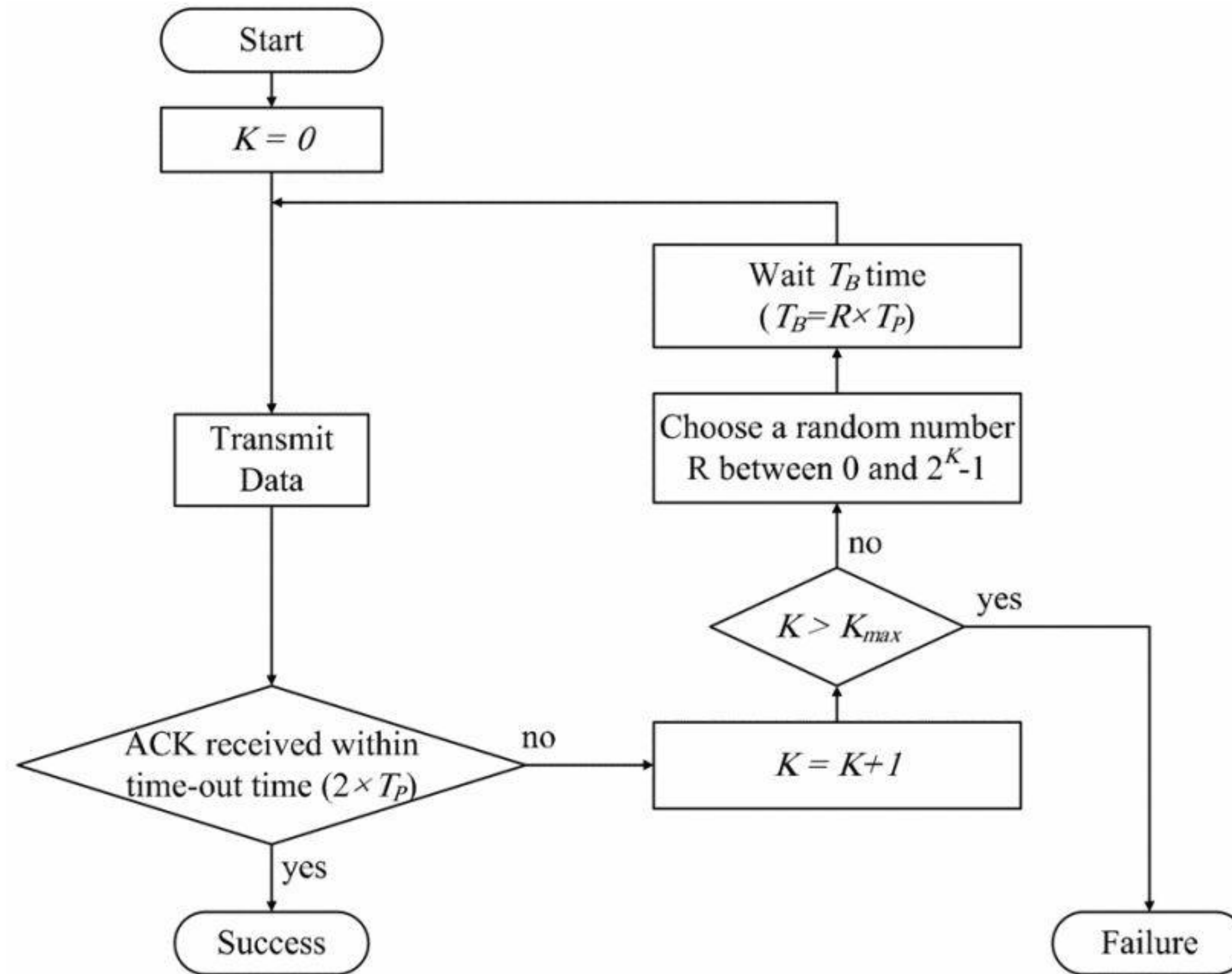
52



Source: Adiono, T. *et al.* MAC Layer Design for Network-Enabled Visible Light Communication Systems Compliant with IEEE 802.15.7. EAI Endorsed Transactions on Energy Web and Information Technologies, 4:14, 2017

IEEE 802.15.7 - Medium access with Pure Aloha

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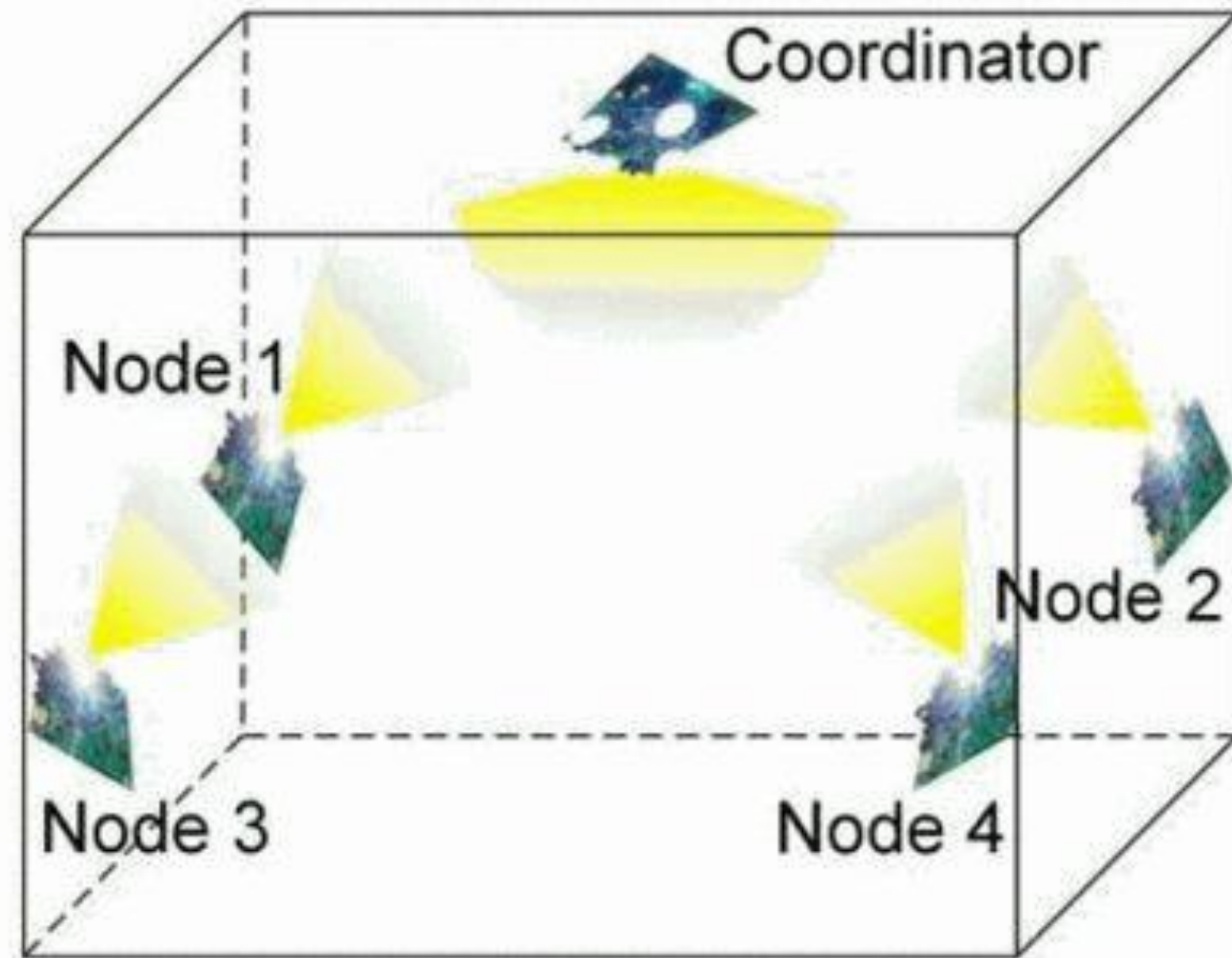


K: transmission attempt
T_p: maximum propagation time
R: Random number

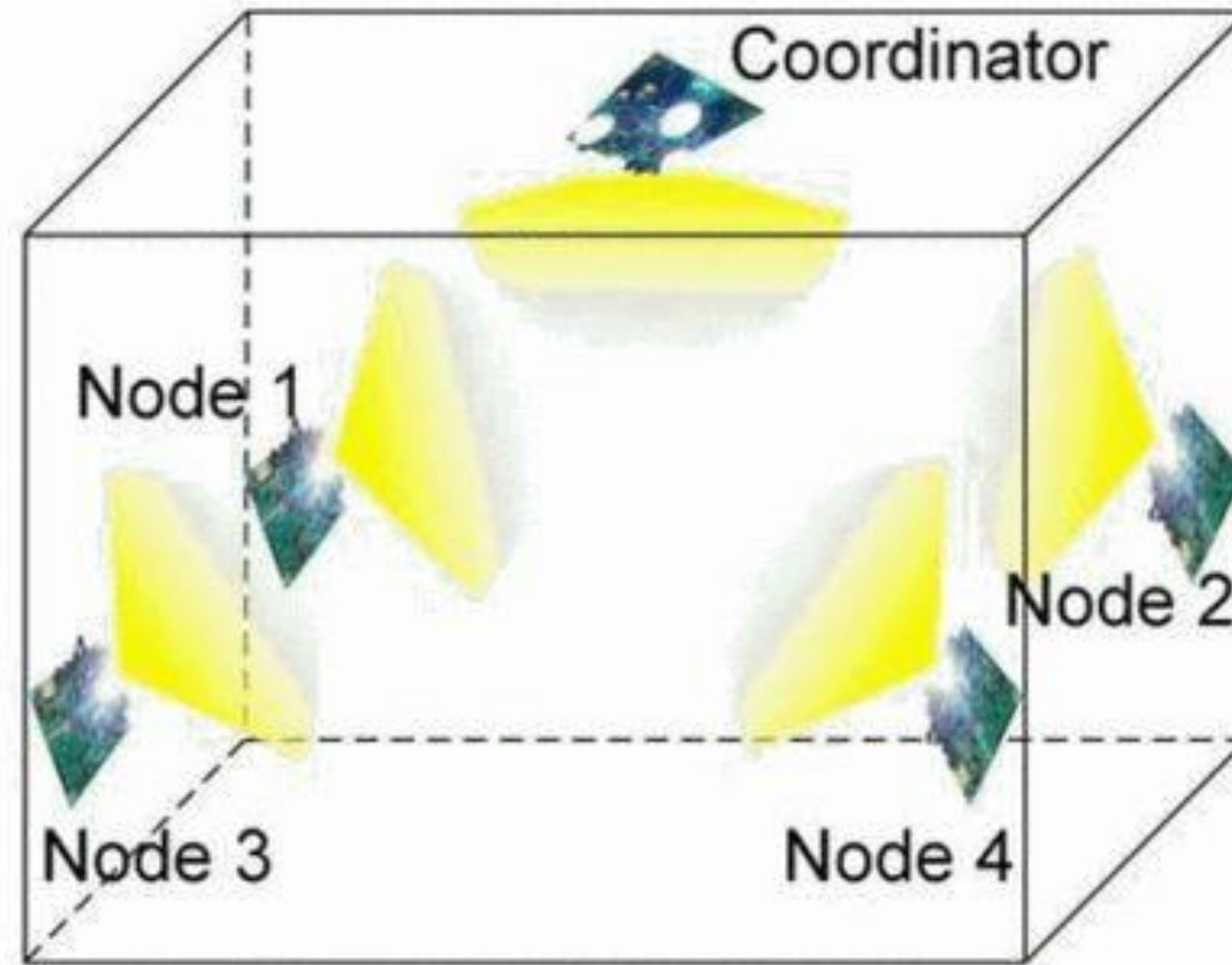
Source: A. Makvandi and Y. S. Kavian, "Experimental Comparison of ALOHA and CSMA/CA for Visible Light Communication IoT Networks," *2022 4th West Asian Symposium on Optical and Millimeter-wave Wireless Communications (WASOWC)*, 2022, pp. 1-4, doi: 10.1109/WASOWC54657.2022.9798436.

IEEE 802.15.7 - MAC Performance

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(a)



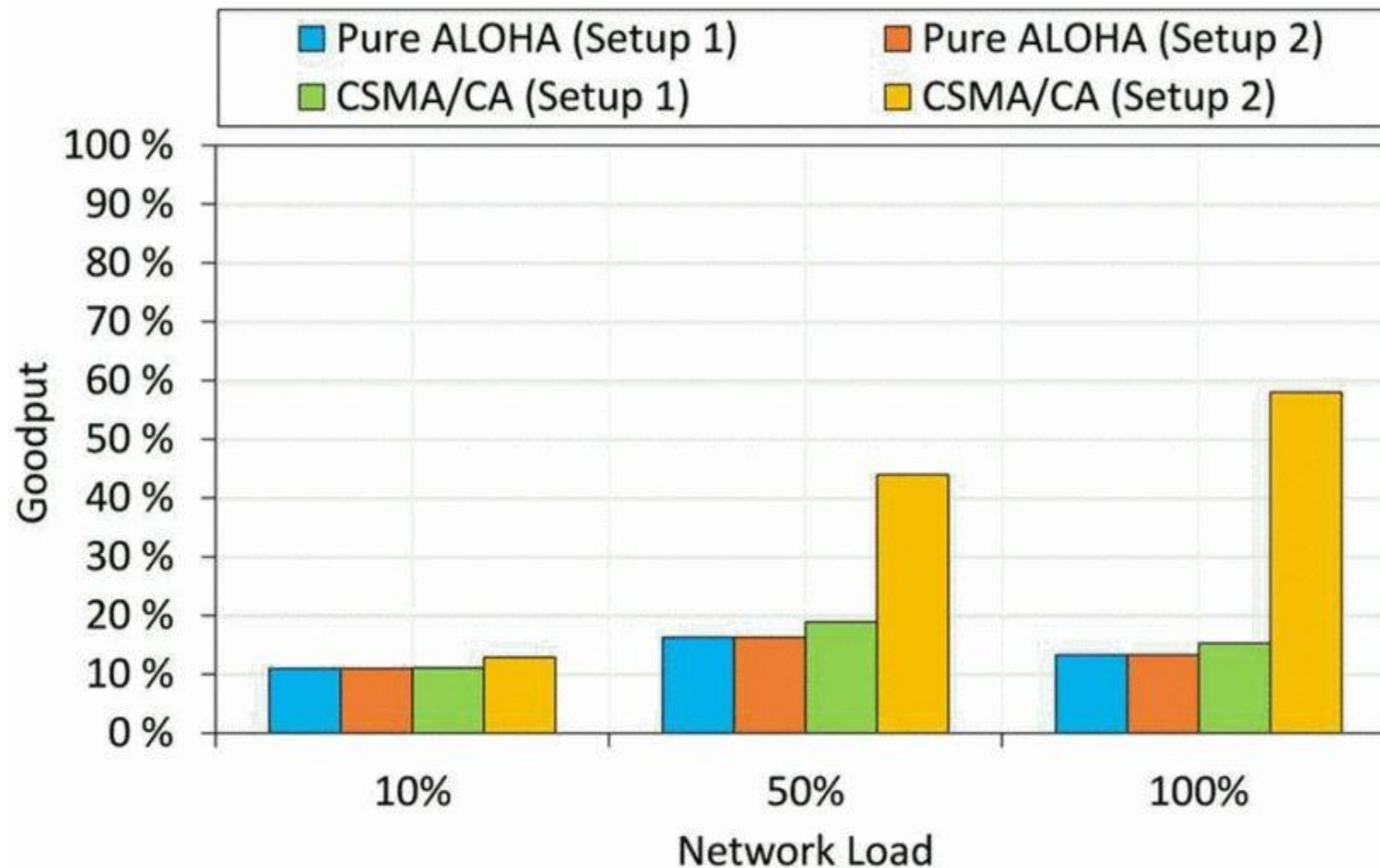
(b)

(a) Setup 1
(b) Setup 2

Source: A. Makvandi and Y. S. Kavian, "Experimental Comparison of ALOHA and CSMA/CA for Visible Light Communication IoT Networks," *2022 4th West Asian Symposium on Optical and Millimeter-wave Wireless Communications (WASOWC)*, 2022, pp. 1-4, doi: 10.1109/WASOWC54657.2022.9798436.

IEEE 802.15.7 - MAC Performance

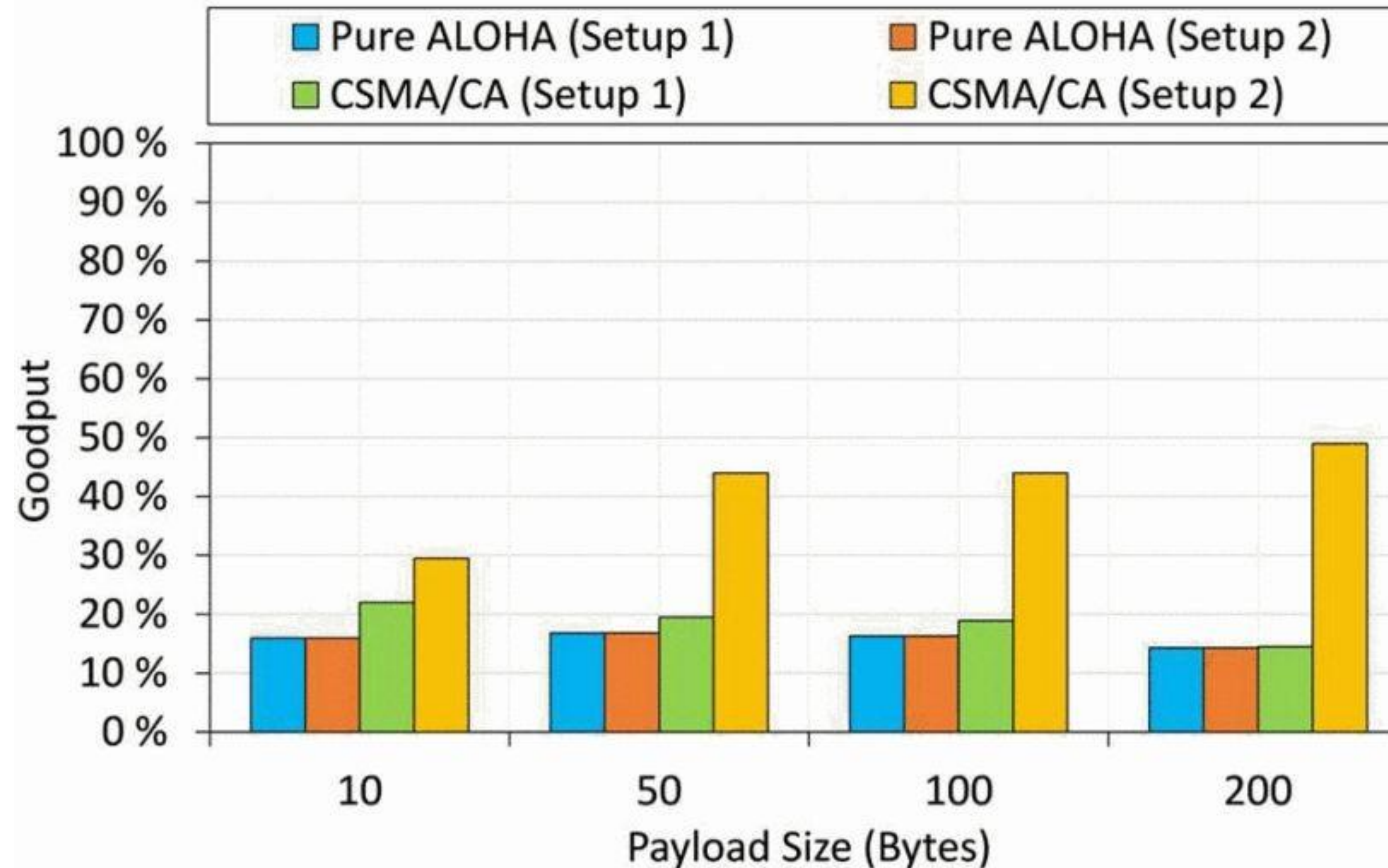
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Source: A. Makvandi and Y. S. Kavian, "Experimental Comparison of ALOHA and CSMA/CA for Visible Light Communication IoT Networks," *2022 4th West Asian Symposium on Optical and Millimeter-wave Wireless Communications (WASOWC)*, 2022, pp. 1-4, doi: 10.1109/WASOWC54657.2022.9798436.

IEEE 802.15.7 - MAC Performance

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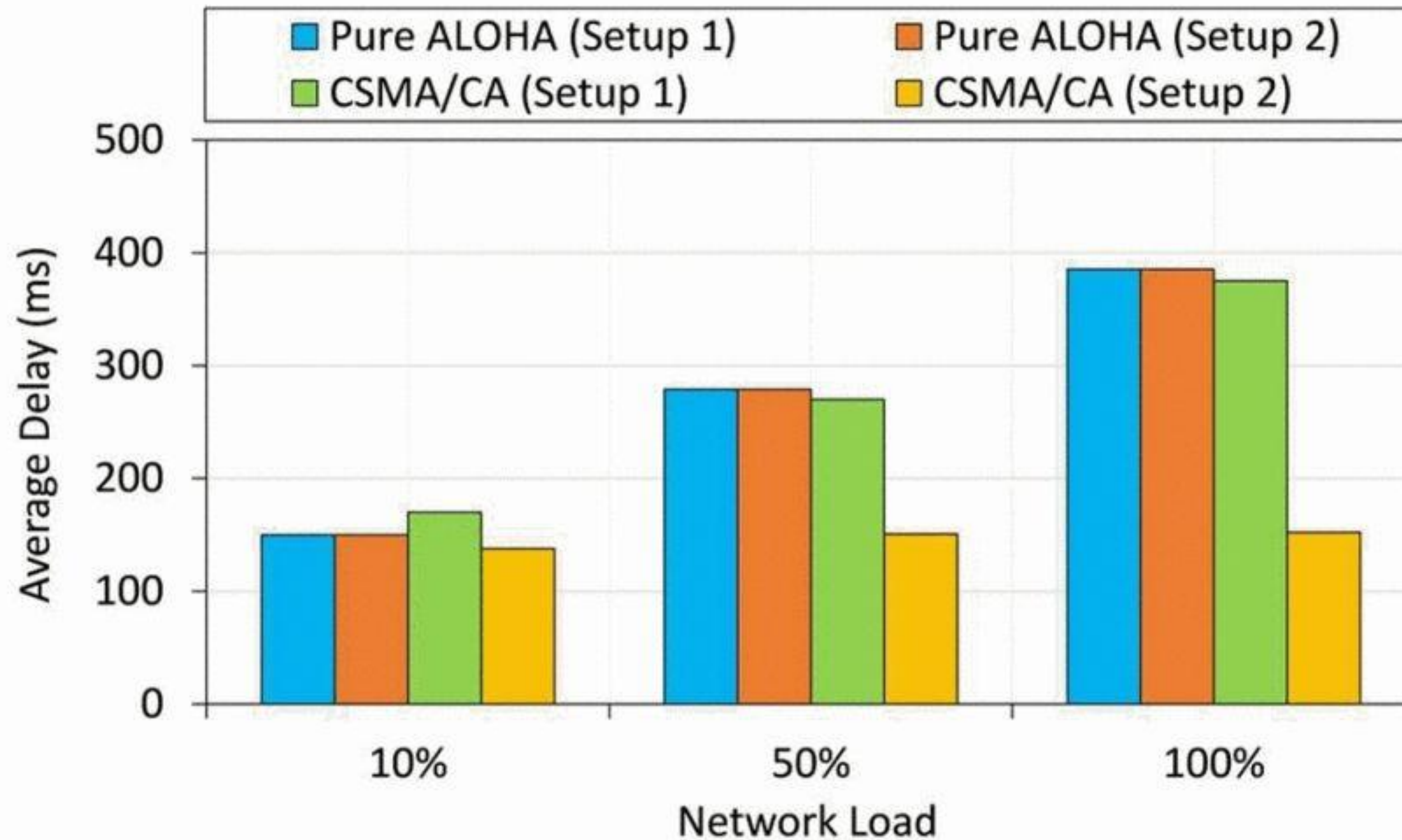


Network load = 50%

Source: A. Makvandi and Y. S. Kavian, "Experimental Comparison of ALOHA and CSMA/CA for Visible Light Communication IoT Networks," *2022 4th West Asian Symposium on Optical and Millimeter-wave Wireless Communications (WASOWC)*, 2022, pp. 1-4, doi: 10.1109/WASOWC54657.2022.9798436.

IEEE 802.15.7 - MAC Performance

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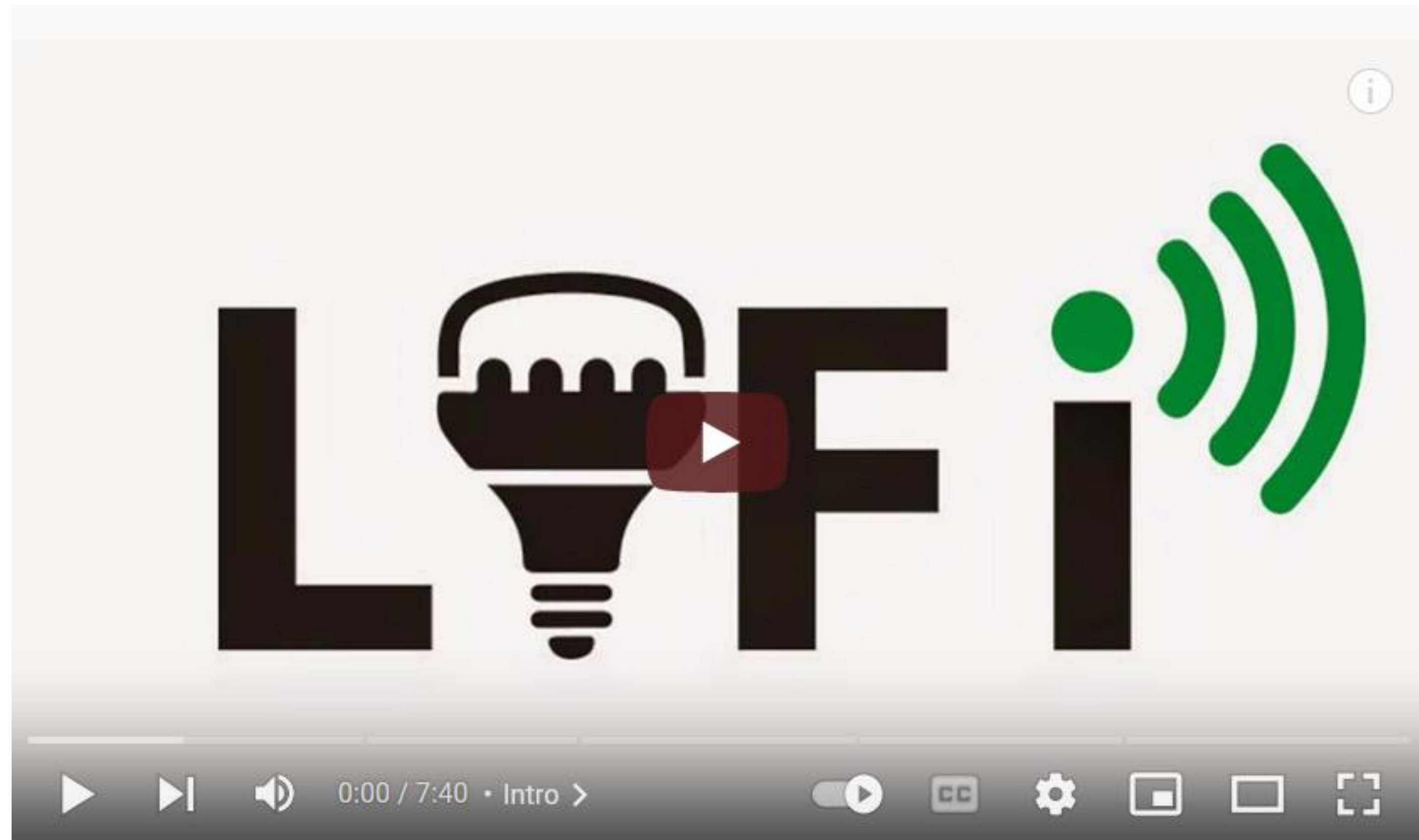


Source: A. Makvandi and Y. S. Kavian, "Experimental Comparison of ALOHA and CSMA/CA for Visible Light Communication IoT Networks," *2022 4th West Asian Symposium on Optical and Millimeter-wave Wireless Communications (WASOWC)*, 2022, pp. 1-4, doi: 10.1109/WASOWC54657.2022.9798436.

LiFi: use-cases

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<https://www.youtube.com/watch?v=wqH9KX9o0vg>



Industry products: <http://vlncomm.com/>

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Conclusion

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- We have reviewed Wireless connection technologies, the base of communications in IoT
- We have included Wireless and mobile networks in the TCP/IP and Internet model
- Wireless sensor networks were presented:
 - Basic concepts review
 - 5G
 - LiFi